SNAP-ing poverty or Stamp-ing on the poor?
The effect of Food Stamps' work requirements on program participation and labor supply

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Abstract

This paper examines the effect of SNAP's ABAWD Work Requirement and General Work Requirement on employment and program participation among older adults. These requirements end abruptly at the ages 50 and 60, and I exploit this discontinuous change in incentives to identify regression discontinuity and difference-in-difference models using the Survey of Income and Program Participation. This is the first paper to estimate the effects of SNAP's work requirements using individual panel data and the first to consider the General Requirement. I largely do not find a significant effect from removing the ABAWD requirement or any relationship between the General Requirement and SNAP participation. However, I find tentative evidence that the removal of the General Requirement reduces employment by 1-5.1 percentage points. This may suggest that the General Requirement may act to increase employment among adults in their late 50s. However, due to several serious limitations, further research is needed to verify these findings before they can be credibly interpreted as the effect of the General Requirement. Additionally, this paper extends existing theory on SNAP's work requirements and identifies several key issues which limit the validity of this topic's literature.

1 Introduction

The Supplemental Nutrition Assistance Program's (SNAP) work requirements are the subject of vigorous disagreement. This debate is saturated with the same ideological tension which characterizes the broader political and economic disputes surrounding work and public assistance. Proponents claim that requiring work reduces poverty by building self sufficiency, or at least discourages dependency by requiring recipients to find a job (Johnson 2023). Critics argue

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that work requirements are punitive, kicking the destitute while they are down and leaving people to go hungry if they lose their jobs, cannot find work, or cannot get enough hours at work. The vindication of either perspective depends, partially, on the answers to two key empirical questions, which this paper seeks to answer: Do SNAP's work requirements cause increased employment? Do they cause a decrease in program participation?

SNAP imposes two distinct sets of nested work requirements: the "Able Bodied Adult Without Dependents (ABAWD) Work Requirement", and the "General Work Requirement". These broadly apply to most households that consist of working age adults who are not disabled and who do not have children. While political and academic debate has largely focused on the ABAWD Requirement, similar arguments could be made about the General Requirement. Indeed, to my knowledge this is the first paper to attempt to identify the causal effects of the General Requirement. As it is usually described, the ABAWD Requirement says that recipients aged 18-49 may not receive SNAP benefits for more than 3 months unless they work at least 20 hours a week, but as we shall see this description is not entirely accurate. The General Work Requirement applies to recipients aged 16-59, including those who are subject to the ABAWD Work Requirement. To simplify, it requires individuals to register for work, not voluntarily quit a job or reduce their work hours below 30 a week, and may sometimes mandate participation in SNAP Employment and Training (E&T). Noncompliance can result in a loss of benefits for a period between 1 month and permanently.

The cessation of these work requirements at age 50 and 60 mean that SNAP recipients and those on the margin of participation face abrupt, and exogenous, changes in incentives at these ages. I exploit this treatment using both regression discontinuity and difference in difference models to answer the stated questions, identifying the effects of the work requirements by observing how recipients behave when they end. In studying the effect of crossing an age cutoff, it is useful to observe individuals over time so as to account for unobserved individual qualities and ensure that any effects are identified on changes within individuals. For this reason, I use data from the 2001, 2004, and 2008 panels of the Survey of Income and Program Participation (SIPP). This rotating panel is well suited to answering my question as it tracks individuals over years, with a specific focus on employment, income, and social welfare recipiency.

Given that the source of variation is an abrupt change in a program rule, I estimate a variety of regression discontinuities in the vicinities of 50 and 60, using both within-variation and between-variation. The credibility of this approach is limited by a variety of issues, particularly dynamic treatment effects and a lack of a comparison group. To partially address this, I estimate difference in difference models based on comparing those who were and were not exempt from work requirements as they turned 50 or 60. Nevertheless, serious threats to validity remain and there are reasons to think that all estimates are biased towards 0. I do not find a clear and significant relationship between program participation and aging out of either of the two work requirements. I cannot say anything meaningful about the ABAWD requirement because of severe sample dilution due to an inability to observe local waivers. I find some evidence that the removal of the General Requirement may reduce employment by 1 to 5.1 percentage points. This suggests that the General Requirement may act to increase employment among adults in their late 50s. These results are robust in DID specifications which estimate the change in employment at 60 among those subject to the requirement compared to those not subject to it. However, due to several serious limitations, further research is needed to verify these findings before they can be confidently interpreted as being the effects of treatment.

This paper contributes to the existing literature in several ways. It extends the approaches of other papers to a new dataset. This is also the first paper to estimate the effects of work requirements using individual panel data. This allows for estimation using individual fixed effects and within-variation. Additionally it extends existing theory on the mechanisms by which SNAP's work requirements may have an effect. This paper also identifies several institutional features which have been overlooked by previous papers. Finally, this is the first paper to attempt to identify the causal effects of the General Requirement [1].

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2 Background

The Supplemental Nutrition Assistance Program (SNAP) is the largest federal means tested near cash public assistance program in the United States, and is intended to help low income households afford food. It is set apart from other major cash and near cash assistance programs by the fact that it is available to individuals regardless of family structure, disability, or age. As of 2023, it provided $ 107 billion in benefits to 42 million monthly recipients (Food and Nutrition Service, 2024). SNAP was called the Food Stamp Program (FSP) prior to 2008, and it is still colloquially referred to as "food stamps". Given that this paper's period of study spans this name change, the cosmetic nature of the distinction, this paper shall discuss SNAP and Food Stamps as though they were a single program, and will use their names interchangeably.

From its inception, SNAP has been a mostly federal program, being overseen centrally by the Food and Nutrition Service (FNS) of the US Department of Agriculture (USDA). The current incarnation of the program was established in 1996 under the Personal Responsibility and Work Opportunity Reconciliation Act, though some of the federal regulations governing implementation of the work requirements were not finalized until 2001. Benefits are federally funded

[1] An NBER working paper on the General Requirement was released during the final stages of editing this paper. See Cook and East 2024.

and most rules and policies governing the program are determined at the federal level. However, "States, through local welfare offices, are responsible for day-to-day operations of the program and determine eligibility, calculate benefits, and issue benefits to participants according to Federal rules." (Economic Research Service, 2024). As a result, eligibility and benefits are, unlike many programs, theoretically mostly uniform across states. This lack of variation may be the reason why SNAP has received less attention from researchers than smaller programs (such as TANF). In practice, variation in administrative policies and capacities on the state and county level mean that the practical logistics of applying for and receiving benefits, as well as verifying eligibility and enforcing compliance, can vary between places (Nesbit, 2024, Higham, 2024).

The central component of SNAP is the provision of monthly food vouchers. Benefit amounts, referred to as allotments, depend primarily on income and household size, though other factors like certain living expenses can also affect them. Allotments make up the difference between 30% of a household's monthly net income, and the amount the household would need to spend on food each month under the USDA's Thrifty Food Plan. Thus, benefits can range from paying the entire amount of the Thrifty Food Plan's projected food cost, down to a token minimum benefit of a few dollars for households whose income is high enough to reduce monthly benefits to 0 but too low to cause them to lose eligibility.

SNAP eligibility depends on a number of factors, foremost of which is income. Generally, to qualify for SNAP benefits, a household must meet 2 income tests. Firstly, a household's gross income must be less than or equal to 130% of the poverty line. Secondly, after certain deductions are applied, the household's remaining income (net income) must be at or below the poverty line (which it usually is). The exception to the gross income test is that individuals are no longer subject to it once they reach the age of 60. This has potential relevance to this study's design which is discussed in a later section. Households must also meet an asset limit, which can vary depending on a large number of factors, including state. Additionally, SNAP eligibility also differs depending on immigration status, as well as several miscellaneous criteria which are not relevant to this study in most cases.

Work requirements are the other main factor in determining eligibility. By default, SNAP imposes work requirements on recipients who are able bodied working age adults without dependents (ABAWDs). SNAP has 2 sets of nested work requirements: the General Work Requirement, and the Able Bodied Adult Without Dependents Work Requirement (also called the time limit work requirement). These requirements are nested, with General Work Requirement also applying to those who are subject to the ABAWD Work Requirement. Thus, with very few exceptions, those under the ABAWD requirement can be considered a proper subset of those subject to the General Requirement.

The ABAWD Work Requirement applies to all recipients aged 18-49 who are not exempt due to: disability, pregnancy, having a dependent under 18 in their household, or being exempt from General Work Requirements (note that the age was increased to 52 in 2023). All such recipients are (in most cases) required to participate in work, volunteering, or job training for 80 hours a month (technically 20 hours per week). Individuals subject to the ABAWD requirement, but not fulfilling it, retain benefits for a 3 month grace period. After this time limit they become SNAP ineligible for 36 months, or until the requirements are fulfilled.

Waivers and exemptions are two key exceptions to the ABAWD Work Requirement. States can submit requests to the FNS that the ABAWD Work Requirement be waived in geographic regions experiencing economic hardship (defined by specific criteria) which would make it hard for individuals to find a job. For example, most of the nation was exempted from ABAWD Work Requirements under these geographic waivers for several years following The Great Recession. Under most circumstances, states apply to the FNS for permission to waive the requirement for a specified period of time within a specified geographic territory. These applications are usually granted and usually 30-100% of the country is waived at any given time (Cuffey et al. 2022)[2]. Waivers are most often applied at the level of the state or county, but they can also be applied to municipalities or within an arbitrary region not continuous with any political boundary. Additionally, each year states receive a number of exemptions equal to 15% of the number of ABAWD SNAP recipients which that state had during the previous year. The state can then use these to exempt program participants from the ABAWD Work Requirements. The criteria used by states to grant these exemptions varies greatly from state to state and year to year, and, in some cases, is not transparently available.

The General Work Requirement has not been extensively studied by past papers and so I shall discuss its provisions in greater detail. It applies to recipients aged 16-59 and requires individuals to register for work, participate in SNAP Employment and Training (E&T) or workfare if required to by their state, accept offers of suitable employment, not voluntarily quit a job without good reason, and not voluntarily reducing work hours below 30 a week, or weekly earnings below 30x the federal hourly minimum wage, without a good reason (United States Code, 2001, Food Stamp and Food Distribution Program, 2001). For various reasons, including definitions, enforcement, and penalties, it remains to be seen whether these requirements have an impact on employment and eligibility for SNAP recipients and potential recipients. These requirements are often vaguely defined and states generally have broad discretion over how most key terms are defined. While there does not appear to be any official FNS information about what "registering for work" involves, several state sources

[2] See Cuffey et al. 2022 for a more extensive discussion of the ABAWD Work Requirement's waivers

indicate that it is a minimally burdensome administrative process which often happens automatically upon an individual's application or recertification, and that, at most, it can involve an applicant filling out some additional paperwork at a DOL office (New York State Office of Temporary and Disability Assistance, 2023, Patterson, 2013). They indicate that work registration exists primarily as a tally used to allocate federal SNAP E&T funding. SNAP E&T is federally funded and state run and is conceptualized as requiring a level of effort "comparable to spending approximately 12 hours a month for two months $\cdots$ making job contacts" (US Code 2001). However, within some very broad guidelines, states have discretion about what E&T involves and who it requires to participate. Under statute, it can involve a wide variety of activities including workfare, job application programs, and job training. States can decide which of these, if any, to offer, what regions to offer it in, and whether it is voluntary or mandatory. Unfortunately, there is very little information on what E&T typically involves or requires [3]. A 2018 GAO report highlighted a lack of "complete and accurate information" on E&T and that, "As a result, USDA's ability to assess whether agency goals are being met through the SNAP E&T program is limited, as is the department's ability to monitor states' implementation of work requirements and ensure program integrity. " (United States Government Accountability Office, 2018). What is more clear is that the number of people participating in SNAP E&T has usually been small and declining. The same report finds that, as of 2016, only about 207,000 individuals participated in E&T in a given month, equivalent to about 3.4% of those subject to the General Requirement, or 0.5% of all SNAP recipients. This is a decline from 2008 when 8.1% of those under the requirement participated in E&T in a given month (0.92% overall). This somewhat conflicts with a report which found that 629,000 individuals participated in E&T in 2013, equal to 4.7% of work registrants, though the discrepancy may be due to different definitions (Rowe, Brown, Estes, 2017).

Thus E&T may not have much material impact for most of those under the General Requirement. The text of the federal rule against "voluntary quits" appears to indicate that it only applies to a head of household who voluntarily quits a job of 20 hours or more per week without good reason, though the entire household will lose eligibility, and is retroactive to 60 days before applying for SNAP (Food Stamp and Food Distribution Program, 2001). However, the details of this rule are convoluted and I was unable to verify whether it also applies in other circumstances. The rule against voluntary reduction of hours/earnings appears to apply only in circumstances where an individual was already working more than 30 hours per week (or earning more than 30x federal minimum wage) (Food Stamp and Food Distribution Program, 2001). Thus, for example, it would not penalize a minimum wage worker for reducing their weekly hours from 25 to 24. "Good cause" is the exception to the rules against voluntarily quitting or reducing hours. It can include a huge number of circumstances

[3] This appears to be a long running issue United States General Accounting Office, 2003

such as illness, lack of childcare or transportation, discrimination, returning to school, working in certain industries, and a job not satisfying the "suitable employment" criteria. Of note for this study, voluntary retirement, which is recognized by the employer as retirement, is considered "good cause". Thus, a 58 year old choosing to retire would not violate the General Work Requirement. The penalty for failing to comply with the General Work Requirement is losing eligibility for a length of time. This may apply to either an individual or their entire household. The length of time depends on state, the requirement violated, and the number of times the requirement was violated but can be anywhere from 1 month to a permanent ban on receiving SNAP benefits. (United States Code, 2001, Food and Nutrition Service, 2018). Unlike the ABAWD Requirement, there is not typically an official grace period between failure to comply and the loss of benefits, though it is unclear how often compliance is typically verified. Taking the various components of the General Requirement together, they may or may not create a meaningful and binding set of incentives for those on and off SNAP, depending on circumstances and location.

Individuals are exempted from the General Requirement if they: already work at least 30 hours a week or earn 30x the federal minimum wage; already meet the work requirements for another program (such as TANF or UI); are caring for a child under 6 or an "incapacitated" person; are disabled; are in substance rehabilitation; or are a student at least half time. From a certain perspective, achieving exemption by having weekly earnings equal to 30x the minimum wage can be thought of as another way to comply with the General Requirement, and so the General Requirement could be viewed as a requirement to earn 30x the minimum wage or else suffer the penalty of being subject to more invasive requirements.

The program as it exists according to the broadest description of its official rules can differ considerably from the program as it exists in practice. Thus the effect of the work requirements, and their influence on the design of this paper, depend on how, and whether, the rules are enforced. For example, while ABAWD are officially required to work 80 hours a month or else be forced to leave the program after 3 months, the nature of the recertification process means that it can take as long as 18 months for an individual's compliance to be reviewed and their benefits rescinded (Gray et al. 2023). Unfortunately, it is difficult to systematically observe the details of SNAP on the ground in thousands of counties in all 50 states across decades, especially since many aspects of the program are poorly documented, and so one must partially rely on broad generalizations and anecdotes. In practice, state and county agencies may have little ability to observe whether recipients are fulfilling the work requirements. For example, how would a county bureaucrat know that an individual declined an offer of employment in violation of the General Requirement (badfordabidness, 2024)[4]? Additionally, some potential behavioral responses to the work

[4] I contacted a SNAP caseworker who was helpful enough to answer my questions. This is

requirements are premised on recipients being aware of the program's rules. Recipients are likely to be unaware of some program features, and, outside the federal statutory code, it is difficult to find official federal sources which document key mechanics of the General Work Requirement . [5]

Past studies have failed to adequately account for several ways in which the two work requirements may interact. Firstly, they fail to fully acknowledge or account for the fact that an individual is exempt from the ABAWD Requirement if they are exempt from the General Requirement. If someone is receiving unemployment benefits (UI), they are exempted from both work requirements (United States Code, 2001, badfordabidness, 2024). Rather than losing food stamps after 3 months due to working less than 20 hours, they can remain on it for the duration of their UI and then for 3 more months after that. Earning 30x the federal minimum wage per week exempts one from the General Requirement, and thus also the ABAWD requirement. Someone earning the minimum wage would still need to work at least 20 hours, but anyone earning more than 1.5 times the federal minimum wage could become exempt from both work requirements while working less than 20 hours per week. All else equal, to the extent that inflation produces nominal wage increases, we would expect that the number of such recipients would increase with the amount of inflation since the last federal minimum wage increase. Additionally, as of time of writing, 30 states had a minimum wage above the federal, and 25 had minimum wages at least 1.5 times the federal minimum wage. Many cities also have higher minimum wages (Department of Labor, 2024). Thus, this mechanism could exempt many people from both work requirements. We would expect the extent to which the ABAWD requirement binds, and thus the effects of adding and removing it, to vary systematically by location, year, and skill level.

Past papers have also failed to consider how the General Work Requirement may reduce the labor response to the end of the ABAWD requirement as it prohibits voluntarily decreasing weekly work hours below 30 and prohibits heads of household from voluntarily quitting jobs of 20 hours or more per week (see theory section for caveats). However, it is unclear whether these restrictions hold in practice as there are a number of loopholes and exceptions to these rules which may still allow for some labor response among incumbents. Theoretically, these limits do not constraint new applicants so long as they quit their jobs before joining SNAP; however, there are several ways in which they may still have an effect. The prohibition on heads of household voluntarily quitting extends to 3 months before they apply for SNAP. Given that most potential applicants will have low income and limited financial resources, it may not be feasible for them to quit and then wait out the 3 months before applying. Similarly, someone

the APA notation for citing communications through online comments

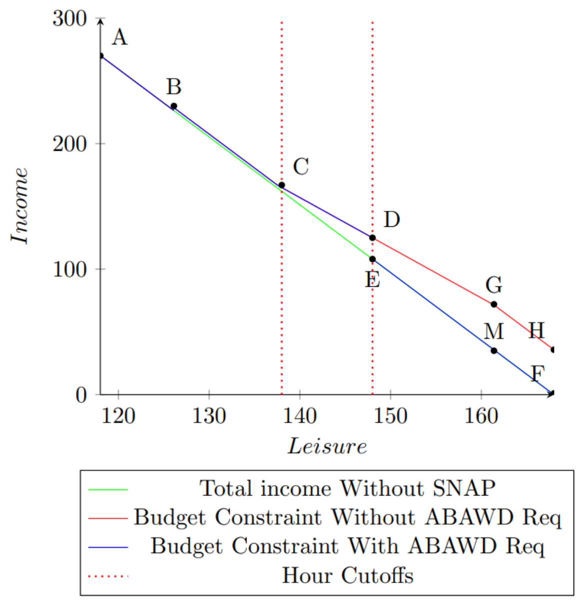
[5]Additionally, There is no guarantee that government agencies always follow the official program rules. For example, under federal statutes (US Code, 2001), benefit agencies are required to inform recipients if they are subject to the General Requirement. However, this did not occur for either of the 2 SNAP recipients who I lived with while writing this paper.

could reduce their work hours before applying, but they would face reduced income while undergoing the application process. Finally, for an incumbent or newly joining participant, the General Requirement requires acceptance of any offer of suitable employment and participation in E&T (if assigned) which can potentially involve applying to jobs. This may limit the extent to which someone can avoid employment while on SNAP.

3 Theory

3.1 Figure 1

1A



1B

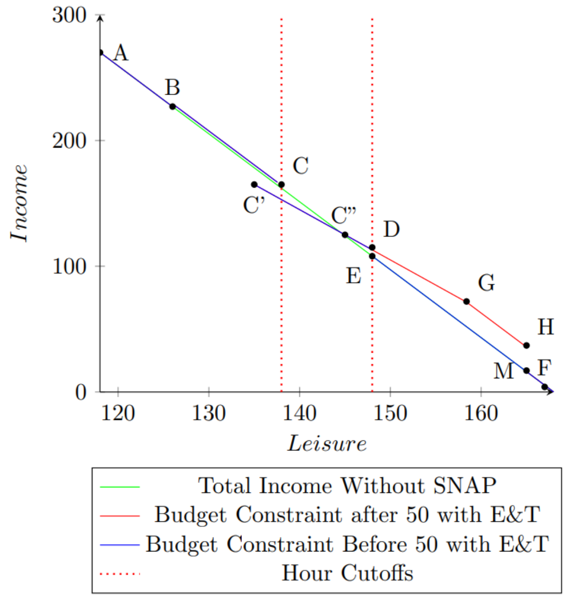


Figure 1 shows two numerically and visually (drawn to scale) accurate budget constraints imposed by the ABAWD Work Requirement on individuals in 2007 in a 1 person household who was earning the federal minimum wage or FMW ($ 5.85), who does not take the SNAP housing deduction, and who is not exempt from either work requirement (graph deducts payroll taxes but not income taxes). Regarding those who are kicked from the program due to non-compliance, this graph depicts a point in time after this has already occurred (already been kicked). Figure 1A depicts the scenario where the General Work Requirement does not require any SNAP E&T hours (or these hours pay minimum wage). Figure 1B depicts a General Requirement to participate in E&T for 3 hours a week, unpaid, for those earning less than 30 times the FMW. As discussed in the Background section the former situation is likely to be far more common. The blue lines represent the budget constraint when both work requirements are in place. The red line, which is contiguous with the blue in most places, represents the budget constraint for someone who is no longer subject to the ABAWD requirement after turning 50. A feature of these graphs which is difficult to represent visually is the General Requirements' "ratchet effect" that constrains how an individual can move along the budgets. Both before and after 50, individuals may not voluntarily move from any point on BC to any point to the right of C without losing coverage. Additionally, even after turning 50, a head of household may not move from D to H without losing coverage [6].

For both 1A and 1B, we would expect that any labor or participation response to lifting the requirement would be driven by people at D and on EMF.

[6] Though, as noted in the Background section, there are many caveats to this

The figures suggest we will observe minimal response from people in regions ABCD for figure 1A, and ABC for 1B (with the possible exception of those at C). We would expect few, if any, individuals to participate in SNAP while on C'C" as they would strictly prefer to be on CC" [7].

Those at point D, referred to as group 1, are program incumbents who are complying with the ABAWD requirement by working 20 hours per week (in 1B only 17 of those hours are paid) but some of whom would have preferred to remain on the program while working and earning less. The General Work Requirement may sometimes constrain their ability to quit a job while remaining on SNAP. For this group, I expect the removal of the ABAWD Requirement to decrease, or not increase, labor supply on the extensive margin if this feature of the General Requirement doesn't bind. Incumbents who were working may quit their jobs in response to the income effect produced by being allowed to keep their benefits while working less. As discussed in Han 2022, the substitution effect from the benefit phase out is likely to have minimal impact on them as they already experience this aspect of the program, and so the change in their incentives experienced at 50 is modest.

For both figures, those in region EMF, referred to as group 2, are non-incumbents who cannot or will not work 20 hours per week while under the work requirement, and who are thus kicked from the program after 3 months. Additionally, those in EMF who join SNAP may quit due to the benefit reduction formula's substitution effect, which is likely to be larger than for incumbents as it is newly imposed on them. However, the presence of a labor substitution effect will depend on where they initially fall on EMF. If they start at EM and move to DG then SNAP's benefit formula will change their real wage and price of leisure, thus causing them to reduce, or not increase, their labor supply. However, as Han points out, if their initial income is low enough then they will go from MF to GH and will not receive the maximum benefit and the slope of their budget constraint would not change, and thus no substitution effect would occur. The income effect would be experienced by both subgroups, increasing leisure and reducing employment. The General Requirement would not constrain this as they were initially working less than 20 hours. Group 2 would experience an unambiguous increase in participation through increased program entry.

For those who were at F when they turned 50, while they have no employment to reduce, receiving SNAP income after the removal of the ABAWD requirement may reduce employment in the long run by reducing job seeking incentives. If they would have found a job in the next month if subject to the work requirement, and now will not do this, then the reduced job seeking will reduce their potential employment outcome going forward. Working incumbents who

[7] Someone may not be able to adjust their hours but they can always move to CC" by quitting SNAP and E&T

choose to continue to work after 50 may respond similarly in the event that they lose their job. However, joining SNAP may also increase long run employment for those at F as it will make them subject to the General Work Requirement which encourages work and provides employment resources which, if successful, may produce increased employment.

In addition to the effects represented in this static model, we would expect the removal of the ABAWD requirement would cause individuals in group 1 (and group 2 after they enter the program) to experience reduced program attrition, and thus higher participation, in the long run, because those meeting the requirement will not lose coverage if they stop meeting it for reasons exogenous to the presence of the requirement, eg: they get laid off and can't find another job, or accidentally under-report their hours. An effect to increase attrition is borne of the interaction between the work requirements. Individuals may quit their jobs, or reduce work hours below 30, in response to being newly freed from the ABAWD Work Requirement, not realizing that this violates the General Work Requirement which they are still subject to and may thus lose SNAP coverage. As is discussed below, all of these effects to reduce attrition are likely to take time, potentially over a year, to fully play out. Finally, incumbents are less likely to work if they are not expected to, all else equal, because humans may be influenced by official rules which tell them what they are "supposed to" do. This goes beyond the incentives of the program itself and includes subtler pressures, such as those exerted by a case worker calling to remind participants to work.

We would not expect a significant labor or participation response from those working a large number of hours on ABCD and ABC on 1A and 1B respectively, referred to as group 4. For Figure 1A, it is difficult to draw an indifference curve satisfying monotonicity which would induce them to prefer to work more than 20 hours under the requirement, but less than 20 hours, without also having them prefer to work an intermediate amount while participating. While such a curve is technically possible, it would require extremely strong restrictions on its form. An individual working x¿20 hours would have to have a near constant rate of substitution between leisure and income (close to the minimum wage) over baskets where they work less than 20 hours. This rate of substitution would also have to be close to the slope of the budget constraint. For example, in order for an individual at B to prefer G, but not prefer a point on CD over B, then their indifference curve would need to be nearly parallel to the budget constraint with a near constant slope between B and G. The same arguments apply to those in Figure 1B who initially work more than 30 hours. While these conclusions hold for the static wage profile in Figure 1, as shown in Appendix Figure A1, this is even more the case for individuals with a higher hourly wage. Figure A1 also shows that we would not expect many of those making a higher wage to lower their wage and work hours while joining SNAP as any indifference curves which would cause this would be a proper subset of the already restrictive indifference curves discussed above. If you are earning significantly more than the minimum wage, meaningful benefit allotments don't kick in until after you are working less than 20 hours. This means that for someone on the margin of reducing their income in order to qualify, they would receive next to 0 benefits unless they reduced their income far below the threshold necessary to qualify. Thus, we would not expect people working more than 20 (or 30) hours to have a labor or participation response to the removal of the ABAWD requirement at 50 unless we are willing to make strong assumptions about their preferences. While there are no doubt some individuals whose preferences satisfy these, it would be a stronger assumption still to assume that there are very many such people. This has significant implications for literature.

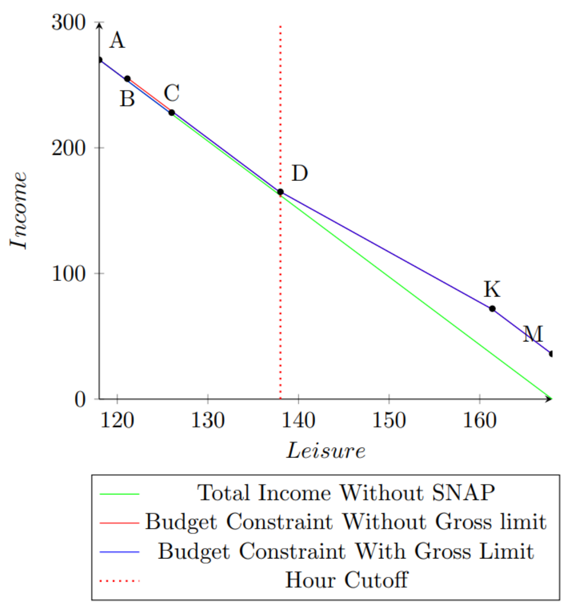
Gray et al. suggest that that many existing studies on the labor supply effects of SNAP work requirements may be severely biased by a failure to account for people lowering their income in order to qualify for SNAP once the work requirement is lifted, leading to selection bias in cross sectional samples construction based on income. In this scenario, an individual will work and earn an income above an eligibility threshold in the presence of a work requirement (and thus will not participate), but, once it is relaxed, the individual will work less to lower their income in order to be eligible to participate (let's call these compilers). If individuals are only observed at a single point in time, and samples were constructed based on having an income below a certain threshold, then this would bias estimated effects on participation and labor supply towards 0 because the higher employment, and lower participation, among compliers will be unobserved before the removal of the work requirement. However, in order for this form of selection to occur, an individual would need to be willing to earn more than 130% of the federal poverty line when a work requirement is present, but would prefer to receive SNAP and work less than 20 hours. As discussed in the previous paragraph, preferences which would produce this require strong assumptions and are likely to be uncommon. Thus, while this form of selection may occur in some cases, it is unlikely to be common enough to be a significant driver of the labor supply and participation responses to the removal of the ABAWD Work Requirement.

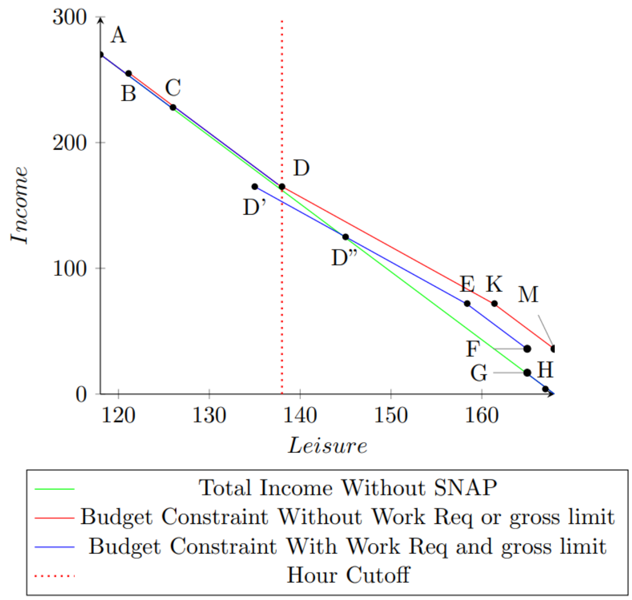
In practical terms, selection bias caused by income is unlikely in most of the studies in this literature as all of them set the threshold for income based inclusion well above the poverty line. Cuffey et al. has the lowest income threshold of 250% FPL. Using 2022 dollars, in order to cause income-based sample selection of a manner suggested by Gray et al. a single individual would need to reduce their monthly income from \$ 3,490 (250% FPL) to \$1,473, a loss of \$ 2,017, in order to qualify. For most such individuals, they would receive a monthly benefit less than \$26. If they sought to further reduce their income in order to gain the maximum benefit then they would need to reduce their monthly income by more than \$3000 in order to gain a total monthly benefit of \$250. They would gain leisure from reducing their work in this way, but this is true regardless of SNAP. While there are no doubt individuals who would make this trade, there are not likely to be very many of them. Thus, income based selection bias of the sort described by Gray et al. 2023 is unlikely to be an issue for this paper, as well as other papers in this literature.

So far we have seen that, in this static model, the removal of the work requirement would affect three groups: incumbents working 20 hours (or slightly above it) located at point D, those not on SNAP who are working less than 20 hours on EMF, and, in the case where there are significant unpaid E&T hours, non-participants on CC". In dynamic terms, this would also affect people who become part of these two groups. If someone with a high income loses their job and is unable to find a new one, then whether there is a work requirement determines whether they can stay on the program and thus determines their long term participation. However, this would not affect their labor supply in the long run. If they were choosing to work and earn more than 130% of the poverty line before they were laid off then their preferences are likely such that they would prefer this over earning less and receiving SNAP, and they are likely to continue to prefer this. The presence or absence of the ABAWD Work Requirement would not induce them to choose to work less but would determine whether they are allowed to collect benefits for more than 3 months.

3.2 Figure 2

2A





2B

Figure 2 shows the budget constraints faced by an individual before and after they age out of the General Requirement at 60, under the same conditions and assumptions as Figure 1. Figure 2A depicts the scenario with no E&T hours and, 2B depicts a General Requirement to participate in E&T for 3 hours a week, unpaid.

In studying the end of the General Work Requirement at 60, a potential confounding factor is that this coincides with the removal of the gross income requirement for eligibility as SNAP participants over the age of 60 only need to meet the net income requirement. However, as can be seen in Figure 2, this will create minimal change in incentives as it would only affect individuals on BC, who will become eligible to receive minimum benefits equal to \$2.30 a week. Given this negligible allotment, I do not expect the removal of the gross income requirement to significantly entice more people to apply, and thus will not likely affect labor supply. However, this might not hold if they take large deductions such as the housing deduction.

The removal of the General Requirements at 60 may slightly increase program participation and reduce labor supply for reasons not depicted in Figure 2. While they also apply to Figure 2B, these are likely to be the only effects on participation among those in Figure 2A, as nothing changes other than the gross income limit. For incumbents, I expect it will increase program participation through reduced attrition as those meeting the requirement will not lose coverage if they unintentionally stop meeting it, for example: by forgetting to show up for E&T, or due to administrative error. It might also reduce long term attrition among those with temporary exemptions such as those receiving unemployment benefits if they would otherwise have lost coverage when those exemptions ended. It is unclear whether easing the General Requirements' restrictions on reducing labor supply would have any effect on participation. Those who wish to work less than 30 hours were always allowed to participate, so long as that reduction in hours occurs before they apply to the program. There might be slightly reduced program attrition among incumbents who would have reduced their hours below 30 or quit their job regardless of the requirement. Similarly, there might be slightly increased program entry among non-incumbents who would prefer to have the flexibility to adjust their labor supply over receiving benefits, assuming that non incumbents are aware of the General Requirement. The fact that retiring is considered a "good cause" to quit may mean that the prohibition on quitting does not bind among those close to 60. Additionally, these restrictions are only likely to be relevant at lower wage rates. As shown in Appendix Figure 2, at higher wages benefits are limited to a token amount for unless individuals already work less than 20 hours. Meanwhile, if the restrictions on working less bind then their removal will produce an unambiguous decrease in labor supply, especially among incumbents.

Several additional effects and considerations apply in the presence of a significant E& T requirement (Figure 2B). We would not expect anyone to prefer to participate if they are working between D' and D", though a few may still participate if they have a limited choice set. Instead, we would expect them to prefer to not participate while on DD", or to participate on CD. When the required E&T hours end at 60 we would expect those on DD" to increase participation and decrease labor supply on the intensive margin due to both the income and substitution effects. For incumbents on D"EF, they will move to DKM and decrease, or not increase, their labor supply. This may occur on either the intensive or extensive margin. There may be some response from those initially on ABCD. However, this is likely to be limited in scale for the same reasons as it is when the ABAWD requirement is removed at 50.

3.3 Asymmetry and Interactions

This paper observes the effects of removing work requirements. However, it is ultimately concerned with a hypothetical comparison of a world with and without work requirements. Meanwhile, in practice, what papers in this literature actually estimate are the (usually) short term effects of adding and removing them. In the case of the ABAWD requirement, we would broadly expect these to be asymmetric in their timing and presentation for several reasons. Holding preferences constant, an individual's behavior has a tendency to be persistent over time. All else equal, a person who is employed today is more likely to be employed tomorrow than a person who is not employed today. (see if I can find anything about this in terms of attachment to the labor force). If the requirement is newly imposed then this individual must increase their labor supply in order to retain benefits. If they age out of work requirements, they are not required to do anything in a particular time span. Thus, for an individual on SNAP who is inclined to choose not to work (or to work less than 20 hours) in the absence of a requirement, but who will comply with the requirement, there is likely to exhibit a lag between the removal of requirements and their reduction in labor supply in response. This lag is likely dependent on an individual's labor force attachment. This would serve to reduce the immediate labor supply effect of aging out of work requirements, while potentially leaving the long term effects unobserved. This would bias a static estimate of labor response towards 0. In contrast, when the AWAD requirement is imposed on an incumbent SNAP recipient, the 3 month time limit creates a deadline by which they must begin working 20 hours. This would serve to speed any change in behavior, making more of the observable labor response occur in the immediate aftermath of the work requirement. In practice the deadline to comply will depend on certification periods, which can vary between and within states and years based on a wide variety of factors. There's evidence to suggest (Gray et al. 2023, Ribar et al. 2010) that this may mean that, for the most part, some eligibility requirements are functionally not enforced before an individual's recertification. Ribar et al. has shown that this can sometimes be 3 months, while Gray et al. shows that the lag between imposition and enforcement can take as long as 18 months. As each of these papers only study a single state, SC and VA respectively, during a short timespan (1996-2005 and 2013-2015 respectively) it is difficult to know which of their results are likely to be more generalizable. The average recertification period has grown longer for SNAP participants as a whole between 2001 and 2013 (Economic Research Service, 2020), but it is unclear whether this is true of the average recertification period for ABAWDs. When combined with 15% exemptions, there may functionally be a time delay between when work requirements are officially imposed and when ABAWDs are required to adhere to them. If time delays are long then this would reduce the extent to which ABAWDs face an immediate incentive to comply with newly imposed work requirements. When making static, before-after comparisons, certification delays would reduce the apparent effect of adding work requirements on labor supply and participation. This wouldn't help if the characteristics of the unemployed differed systematically between places which do and don't receive treatment.

The observable participation response to adding and removing the ABAWD requirement is also likely to be asymmetric. The observable participation response to adding and removing the ABAWD requirement is also likely to be asymmetric. The effects of imposing the work requirement is mechanical in the sense that individuals who fail to comply will be kicked off the program. In contrast, the removal of the requirement does not mean that newly eligible individuals will be automatically signed up for SNAP, or even informed of their eligibility. Even if they apply, the application process will take time. All else equal, these factors would make the effect of removing work requirements to be more gradual than the effect of adding them. A static comparison of before and after the work requirement is removed, such as the ones in the present study, are likely to underestimate the participation response. This also applies to removing the General Work Requirement. If longer recertification periods delay the mechanical effects of imposing the ABAWD requirement, this would also lead to underestimating the participation effects. The effects of imposing the work requirement is mechanical in the sense that individuals who fail to comply will be kicked off the program. In contrast, the removal of the requirement does not mean that newly eligible individuals will be automatically signed up for SNAP, or even informed of their eligibility. Even if they apply, the application process will take time. All else equal, these factors would make the effect of removing work requirements to be more gradual than the effect of adding them. A static comparison of before and after the work requirement is removed, such as the ones in the present study, are likely to underestimate the participation response. This also applies to removing the General Work Requirement. If longer recertification periods delay the mechanical effects of imposing the ABAWD requirement, this would also lead to underestimating the participation effects.

One detail which past papers have overlooked is the potential interaction between the ABAWDs and General Work Requirement, which would tend to create a ratchet effect on labor supply. This would constrain the labor supply response to removing the ABAWD Requirement but not the response to imposing it. Several past papers have looked at whether the removal of a work requirement corresponds to a reduction in work hours due to no longer having to meet the 20 hour minimum. However, individuals subject to the General Requirement are forbidden from voluntarily reducing work hours below 30 per week and heads of household are forbidden from voluntarily quitting a job of at least 20 hours per week. This means that, for individuals working over 30 hours when the ABAWD requirement is removed, they may not reduce their intensive margin labor supply to the extent that their work hours would fall below 30. Those working under 30 hours would not be subject to this constraint. Additionally, heads of household who would otherwise quit their jobs with the cessation of the ABAWD requirement cannot do so without violating the General Work Requirement, constraining the extensive margin response. This conflicts with the findings of Cuffey et al 2022, Han 2022, and Harris 2021. Nevertheless, aging out of the ABAWD requirement does still loosen eligibility so that people who were already not working, or were already working less than 20 hours, can now qualify for SNAP. Thus, we might expect reductions in the average work hours among recipients to be driven by individuals joining the program, and not because people are reducing their hours of work. Whether the General Requirement will constrain the response to the removal of the ABAWDs Requirement in practice remains to be seen. A key determinant of any interaction will be the extent to which this aspect of the General Requirement is enforced. It may be very difficult in practice for a program administrator to determine whether or not an individual's reduction in work hours was voluntary. While there are certainly exceptions which would allow an individual to reduce hours or voluntarily quit without running afoul of the General Requirement, we would still expect that they would have some effect to constrain the labor response to removing the ABAWD requirement. While not likely to cause bias, this would mean that the labor response to removing work requirements is limited by this ratchet effect, while the labor response to adding them has no such limitation.

3.4 Discussion of biases

Even in an ideal sample there are likely to be several sources of bias when estimating the effects of aging out of either work requirement. Specific to this and other regression discontinuity papers, Lee and Lemieux 2010 suggest that RD designs with inevitable treatment and age as a running variable may be biased by anticipatory effects. For example, individuals may anticipate the end of the req and quit early, making employment appear lower just before the cutoff. This would reduce the apparent decrease in employment after the cutoff, biasing estimates towards 0. This would likewise bias the estimated effects on participation.

Many of the effects outlined above are likely to take time to materialize, such as reduced attrition, increased program entry, reduced labor supply, etc. A static comparison of those just above, or just below, the age of 50 (or 60) will be biased towards 0 because it will only capture the immediate effects, while failing to capture the long-term effects. Thus, such a comparison should be interpreted as the short-term effects of removing the relevant work requirement, rather than the total effect, and thus should be considered a lower bound.Additionally the effect of reduced attrition manifests when someone goes from being kicked off next month to not being kicked, thus resulting average participation to be higher than it would have been otherwise. If someone is going to be kicked in the next month then this effect would happen in the next month. If, among those at risk of being kicked, someone's probability of being kicked in the next month was $p < 1$ then the full effect to increase participation by reducing attrition will not appear immediately but will take at least $1/p$ months (or probably longer). Thus, the full impact of reduced attrition will take time to occur. As past papers have shown (Ribar 2010, Gray et al. 2023) the ABAWD requirement is most often enforced during an individual's recertification, which can occur at any interval between a month and 18 months. This may mean that de facto observation and enforcement of compliance with the ABAWD requirement can sometimes be infrequent. While the General Requirement is not as well studied, it seems plausible that enforcement would be similarly infrequent. Additionally, even when someone gains eligibility for SNAP by turning 50 or 60, they are unlikely to go out and apply on their birthday. The time lag between gaining eligibility and receiving benefits means that any effect of increased program entry may also take months to occur. Given that these effects to increase program participation will take months to occur, a simple comparison of means on either side of the cutoff could bias the estimated increase downwards towards 0. Any corresponding effects on employment would likewise be biased.

Several sources of bias arise from other changes in participation which may occur at the ages of 50 or 60. It would cause bias if SNAP outreach changes discontinuously, either increasing or decreasing the targeting those over 60. If SNAP outreach increases when someone turns 60 then they will be more likely to apply. This would increase participation by increasing program entry on its own. This would bias my estimates of the effect of aging out upwards, globally, by making it look like the effect of increased outreach is being caused by aging out of work requirements. Additionally, SNAP's recertification periods depend on many factors, including income and age, and it is a reasonable possibility that the number of months between recertifications might increase discontinuously at the age of 50. For someone at risk of being kicked, either due to not satisfying the req, administrative error, or some other reason, this could immediately reduce the number of people being recertified right after the age cutoff, and thus reduce the number of people being kicked right after turning 50 or 60. Thus, concurrently to the end of either work requirement, a change in recertification period would reduce attrition overall, and this increase participation on its own. This would bias my estimates of the effect of aging out upwards, globally, by making it look like the effect of changing recertification periods is being caused by aging out of work requirements. This would probably also interact in some unknown way with the effects that function through program attrition. This cannot be solved by controlling for recertification period [8]. In a sense, this can be interpreted as just an extension of the work requirement and thus changes caused by it could be considered as part of the treatment effect and thus not bias at all. In a similar vein, potential discontinuous changes in SNAP E&T could be seen as a source of bias, but may be better understood as a component of the General Requirement itself. Therefore its removal should be considered a component of the treatment of aging out of the requirement. (see Appendix Section 2 for more explanation).

Another source of bias comes from the fact that, among those who were already eligible but not on SNAP, turning 60 may change people's perceptions in a way that makes them more willing to apply. They may see themselves as deserving or needing it more. The program may have some stigma/baggage related to thinking that it is only for those who "deserve" it. After turning 60 someone might think "I'm old now, so I do deserve it". This would increase participation by increasing program entry on its own. This would bias my estimates of the effect of aging out upwards, globally, by making it look like the effect of changing perceptions is being caused by aging out of work requirements. Additionally, 60 is generally considered a "big" birthday, and so turning 60 may lead people to reevaluate their lives and make changes which might affect participation in ways that are hard to predict. Another source of bias may arise if pensions or other retirement related things change discontinuously at the age of 60. There's no major changes in other federal programs at this age but there could be some change in smaller programs or private policies (employer things, retirement accounts etc). This could increase someone's income and make them ineligible or make them decide that they no longer need it. This would bias my

[8] I don't have the necessary data and it would be a "bad control"

estimates of the effect of aging out downwards globally, by reducing program entry and increasing attrition.

4 Literature review

To my knowledge, there are no papers about the effects of the General Work Requirement at time of writing. Meanwhile, there are several papers which seek to understand the labor supply and participation impacts of the ABAWD Work Requirement. Existing research, and its findings, are limited by econometric errors and a failure to consider timing, asymmetry, interaction with other features of the program, and other issues which I discussed in my theory section. Dynamic treatment effects bias most papers towards 0, while several papers dealing with an age cutoff at 50 are likely similarly biased by anticipatory responses (Cuffey et al. 2022, Han 2022, Ritter 2018).

With some exception of Ritter 2018, existing papers have not discussed how asymmetry and timing may bias the estimated effects of the ABAWD Work Requirement and lead to inconsistent results. We would broadly expect these to differ in their timing and presentation for several reasons. As discussed previously, both would be biased towards 0 by dynamic treatment effects, and these will likely be driven by different factors and occurring over different spans of time. The imposition of requirements may have a larger effect as it acts with a stick rather than a carrot, mandating a change in behavior. Additionally, interactions with the General Requirement mean that we might expect a larger labor supply response from adding work requirements than removing them.

In addition, existing papers are frequently attenuated due to sample dilution. Those exempt from the General Requirement due to taking care of a disabled family member or receiving unemployment benefits are also exempt from the ABAWD requirement. In this case of UI, it seems likely that this would have some form of relationship with treatment and in some way interact with potential outcomes, though the exact nature of this relationship is a topic for future research. Regardless of their specific effects, the role of General Requirement exemptions is an institutional detail that past research has not acknowledged. Additionally, most papers in this literature may suffer from sample selection issues which I discuss in further detail in my Data section.

Different papers have alternately looked at the effects of adding or removing work requirements on different outcomes for different populations. As discussed previously, we would expect the effects of adding and removing work requirements to be distinct. Cuffey et al. and Ritter estimate the labor supply effects of removing the ABAWD Work Requirement at the age of 50. Both study a CPS sample of individuals without a highschool diploma. Ritter 2018 studies individuals between 2000 and 2016 while Cuffey et al. 2022 studies individuals from 2005 to 2009 and further restricts their sample to those under 250% of the federal poverty line. Neither are able to observe disability and so individuals with disabilities are present in their CPS samples, diluting any effect. Ritter also uses samples of the SNAP QC data from 2003-2017 and 2012-2017 containing individuals who did not have children, and did not have children or a disability (respectively). While Ritter does not find any impact on the probability of working at least 20 hours per week, Cuffey et al. find that turning 50 in unwaived and non-exempt area causes "a 13 percentage point reduction in the probability of being employed, a reduction of 4.66 hours worked per week, and a 13 percentage point reduction in the probability that an individual works 20 or more hours per week." These effects are strongest in states with low minimum wage or which provide work to all or most ABAWDs. While both studies attempt to restrict themselves to those not subject to a waiver or exemption, Cuffey et al. do this with far more precision, applying detailed information to identify county-months where nobody was waived or exempt, while Ritter excludes based on having a higher than 5% waiver or exemption usage rate at the state-year level.

Both of these papers have significant problems which limit their credibility. Data quality issues limit their ability to observe key variables precisely, likely leading to dilution. Both fail to acknowledge or account for the panel nature of their CPS data, and thus they estimate "pooled" regression which are at risk of omitted variable bias and type 1 errors. Ritter's remaining regression relies on SNAP QC data which can be unreliable for observing employment, not least because it is drawn from SNAP's official records. If there is any circumstance where people have an incentive to lie about their hours worked, it would be when your SNAP benefits depend on it. For these and other reasons, many of which the author acknowledges, Ritter's results are likely uninformative. Meanwhile, Cuffey et al.'s effect sizes seem surprisingly large given the relatively small proportion of their target population who participate in SNAP and the number of potential sources of downwards bias. Due to the precision and robustness of their estimates and the overall rigor of their approach it seems likely that they demonstrate a real effect of turning 50 to decrease labor supply, however, the magnitude of these effects is suspect.

Several papers look specifically at the effects of imposing work requirements. Ku et al. examine the effect that adding work requirements has on county participation levels between 2013-2017. Meanwhile, in their primary specifications, Harris et al. estimate the effect of adding work requirements on labor supply and participation in a 2010-2017 CPS sample which excludes the disabled and those living with a child. Notably, Harris doesn't restrict by income or education. Reweighs based on characteristics to look like QC ABADs. While they both technically estimate the average difference in outcomes with and without work requirements, rather than specifically the imposition of requirements, their periods of study mean that, in practice, variation in work requirements comes from the removal of waivers. Gray, on the other hand, estimates the effect of imposing work requirements in Virginia between 2013 and 2015 on labor supply and participation among incumbent SNAP recipients, estimated by comparing the change out outcomes among those under and over the age of 50. Harris et al. find that adding work requirements increases employment by 1.3 pp (1.8%) and increases the probability of working 20 hours or more by 1.2 pp, though the magnitude of these effects is difficult to interpret. Gray finds that adding work requirements decreases participation among incumbents by 23.4 pp (-37%), and that this effect was strongest among economically vulnerable populations, while Ku et al. find a county level decrease of Decrease 3-4.5 pp (about -33% of all ABAWDs) in the quarter after work requirements are reimposed. Meanwhile, Gray et al. do not observe an effect on earnings.

Gray et al.'s results are likely biased towards 0, and may have hypothesis testing errors. Additionally, their lack of an effect on earnings may result from the ratchet effects of the General Requirement being particularly relevant to their population of study. Harris has serious issues. Their sample construction is based on absurd assumptions, though their results are somewhat robust to relaxing these. They are definitely biased downwards.

Both Han and Harris (in some specifications) estimate the combined effects of geographic waivers and turning 50 in an ACS sample. These can be thought of both as the additional effect of a waiver on those under 50, or as the additional effect of being under 50 which comes from having/lacking a work requirement. Due to the timing of their periods of study, variation in local work requirements comes from the imposition of work requirements in Harris, while Han uses both imposition and removal of work requirements. Han estimates these effects in a 2005-2017 sample of older adults who were below 300% of FPL, weren't disabled, and didn't live with disabled, elderly, or kids. I do not list Han's estimates here because, while they are very robust, I am doubtful of their validity due to a large number of issues. These include several kinds of random and nonrandom measurement error in both dependent and independent variables, sample dilution, sample selection, and bad controls. Additionally, like several other papers, it suffers from an inability to account for dynamic treatment effects (Goodman-Bacon issues), anticipatory effects, and sample selection due to changes to disability status after the age of 50. See Appendix Section 4, for more details.

Ribar provides detailed descriptive information on the mechanisms through which work requirements affect participation in South Carolina 1996-2005. They provide strong evidence that most program exits occur at recertification (which is also supported by Gray et al.), and that many people are kicked from the program due to the ABAWD Work Requirement time limit "The timing of people's exits lines up extraordinarily well with when the policies should have had their effects, and the associations are robust to alternative comparisons."

From this, there seems to be credible existing evidence that imposing work requirements causes a large decrease in participation, though this conclusion needs to be further substantiated. There is some evidence that adding work requirements may increase labor supply but the validity of these results is questionable. Aging out of work requirements likely decreases labor supply, though it is hard to clearly say by how much. Other than Gray et al. most studies focus on the combined effects on incumbents and non-incumbents. Overall, most existing papers (Cuffey et al. 2022, Gray et al. 2023, Han 2022, Harris 2021, Ritter 2018) in some way involve or focus on the age of 50, which limits what they can tell us about the effect of the ABAWD requirement on other age groups. There are also several methodological limitations which this literature needs to better address and all the studies have significant problems. In particular, they fail to account for the dynamic nature of treatment effects and anticipatory effects, though this is likely to be less of a problem in Ku et al. (2019).

5 Data

I attempt to be the first study in this literature which provides estimates identified on within-variation, observing people over time and comparing individuals to themselves before and after the work requirements are removed. There are several reasons why this is worth doing, which I discuss in greater depth below. It allows me to use individuals as counterfactuals for themselves, providing a clearer source of variation. Additionally, through use of individual fixed effects I can increase the precision of my estimates. This requires that I observe people over time and that I use individual fixed effects to ensure that I am identified based on within variation.[9]

My goal is to estimate the effect of removing the ABAWD and General Work Requirements by comparing people below and above the ages at which these are removed. Several past studies have been identified by individual between variation, using pooled cross sections to compare 49 year-olds to 51 year olds, including observations with characteristics which make SNAP relevant, such as low income, while excluding observations which are exempt from the work requirement. (While Gray et al. do observe individuals over time, they still use between variation. Other studies have done things at the county level). However, in these cross-sectional studies with one observation per person, the average outcomes with and without work requirements are based on comparing 2 different groups of people, who may or may not have similar potential

[9] I also chose the panel data because I originally had the goal to advance the literature by performing secondary analyses which separately examine program incumbents and new entrants. This would have allowed me to determine whether any effects on employment among SNAP participants were due to a change in program composition or a change in the behavior among existing participants. It could have also revealed whether changes in participation were due to new program entry or lower attrition. This is something no other paper has really been able to do because they don't have panel data that allows them to observe people over time. Unfortunately, small sample sizes prevented this.

outcomes and unobservable characteristics. These groups may be poor counterfactuals for each-other if characteristics which determine inclusion in the sample, such as family, disability, or income, vary systematically between the groups. In particular, these characteristics may vary non randomly as people age. In data which only observes each individual once, this will make the subsamples above and below 50 be reflective of different populations. If this selection is correlated with potential outcomes then this will bias results. Though this may occur though many mechanisms, I shall illustrate this concept with a simplified example. Imagine a subpopulation of people who will not participate in SNAP when subject to the ABAWD requirement but who will participate if it is removed. For them, the true treatment effect of removing the work requirement is 100%. Now imagine that, regardless of the requirement, 20% of them would have become disabled at some point after 50, and thus 20% of them would have participated after 50 even if the work requirement were still in place. Now their true treatment effect is an 80%, which can be recovered by estimating the effect of turning 50 while controlling for disability (with its effects identified based on the rest of the sample). Supposing that the age at which someone is observed is random, then cross sectional designs (such as Cuffey et al. 2022) which exclude those who are exempt from the work requirement will drop individuals from this 20%, but only those who are observed after 50, while including the ones who are observed before they become disabled. They will observe 0% SNAP participation before 50 and 100% participation on the right, and will not be able to control for disability. Thus the treatment effect will appear to be 100% treatment effect, and thus their estimates will be biased upwards. While this example is oversimplified, it is likely that some form of this bias will occur, given that disability increases steeply with age during the late 40s and early 50s.

My primary data consists of the public use core microdata files for the 2001 and 2004 panels. Samples pertaining to the General Requirement also include the 2008 SIPP panel. The ABAWD samples do not include this panel because either the entire country, or the vast majority of it, were under ABAWD requirement waivers for nearly the entirety of the period it covers (May 2008- November 2013) (Cuffey et al., 2022, Harris, 2021). Each SIPP panel samples households during its first wave. It then follows the individuals in these households for several years, and adds any members of any household which an original individual may come to occupy. Individuals in the panel are interviewed once every 4 months (or waive) about each of the intervening months. Records in the SIPP data consist of person-months, though all records contain information pertaining to the household, family, and subfamily to which a person currently belongs, such as the household's income, or whether the family contains any children. As such, the person-month is the unit of analysis for this study.

SIPP is well suited to this study as it tracks a large number of individuals over multiple years, with a specific focus on employment, income, and social welfare recipiency. Additionally, while several past studies have struggled with systematic underreporting of SNAP recipiency in (Cuffey et al., 2023, Han, 2022, Ritter, 2018), several papers have found that SIPP does a better job recording program participation (Czajka and Denmead, 2008, Meyer, Mok, & Sullivan 2009). Several errors identified by SIPP's user notes needed to be corrected. I dropped all wave 6 records for household 11, sample unit 916344451699, in the 2004 panel. I re-coded the education variable for certain education levels for the 2004 panel. The 2004 and 2008 panels had editing discrepancies which led to inconsistent or illogical changes in education, sex, race, and birth date. Note that only the SIPP user files for 2004 mentioned these discrepancies. I defined a record as having an age, race, or sex discrepancy if its birthdate, race, or sex was different from that individual's modal birthdate, race, or sex. Out of about 3.8 million initial observations in the 2004 panel, 250,062 observations, from 28,174 individuals, had discrepancies in age across records. Out of about 5.3 million initial observations in the 2008 panel, 140,256 observations, from 10,914 individuals, have discrepancies in age across records. Discrepancies in education, sex, and race were much less common. I recoded birth month (and thus age) as an individual's mode birth month for all records. I kept individuals if they had few age discrepancies ($\leq 30\%$ of their observations had discrepancies) or if they mad a moderate number of age discrepancies ($30\% <$ and $< 50\%$ of their observations had age discrepancies and their mean discrepancy had an absolute value of less than 1.5 months, indicating that the discrepancies were roughly centered on the mode). All other individuals with age discrepancies were dropped. I recoded an individual's race and sex to be their modal race and sex if $\leq 40\%$ of their observations had discrepancies. I created a new level of the race and sex variables and assigned these values to all of an individual's records if more than 40% of their records had discrepancies in race or sex. I define a record as having an education discrepancy if it indicates that someone's highest attained education level had decreased from the first month in which their education was observed. For the purposes of identifying someone with low education (see discussion section) I used the education level from the first month in which an individual's education was observed. Based on several other user notes, as well as examination of the data, there is measurement error in SIPP's measurement of income and employment which cannot feasibly be corrected.

I study 2 work requirements affecting different groups of people, which end at different ages. Thus I construct separate samples, with non-overlapping age ranges, for each work requirement. Samples intended for studying the General Requirement are focused on ages around 60 while those intended for the ABAWD requirement are focused on the age of 50. However, for the sake of brevity, I shall sometimes describe both sets of samples in terms relating to the ABAWD Work Requirement, such as using an explanation which only refers to the age of 50 rather than also separately describing an analogous concept in terms 60. Unless otherwise stated, assume that descriptions in this section apply to both work requirements and that descriptions of one will also apply to the other based on its analogous age range.

As was discussed by Ritter 2018, a key difficulty when considering sample construction for SNAP's work requirements is that one faces a trade-off between bias due to sample selection and bias due to sample dilution. Including too many individuals for whom a work requirement is not relevant dilutes the sample, attenuating estimates, while attempts to restrict a sample to only those for whom it is relevant will likely also drop some of those for whom it would be relevant, thus potentially systematically selecting for certain patterns of potential outcomes. In addition, working with the SIPP means that one must also contend with the need to have a large enough sample to produce precise estimates and avoid inadvertently sample selecting based on the idiosyncrasies of the survey. Finally, I must choose the specifications which are most appropriate to using the variation in each sample. Similar to Ritter 2018 I attempt to balance these issues by estimating a variety of specifications in several different samples, in the hopes their strengths and weaknesses complement one another.

I construct 2 types of samples, "stock" and "combined". Ideally, I would have a large and balanced panel of only those individuals who were exclusively subject to each work requirement before crossing a relevant age threshold, and who I observed for several years both before and afterwards. The "stock samples" are closest to the ideal sample as they consist of a population of people who are observed while below the age cutoff and who are then followed over time as they age across the treatment threshold. For the General Requirement, these samples consist of all person-months between ages 58.5 (inclusive)-61.5 (not inclusive) for individuals who were observed below the age of 60 and who met the other criteria for inclusion in the relevant sample. For the ABAWD requirement, these samples consist of all person-months between ages 48.5(inclusive)-51.5 (not inclusive) for individuals who were observed below the age of 50 and who met the other criteria for inclusion in the relevant sample. The stock samples reduce dilution by only focusing on individuals in a population of interest, are ideal for estimation using within-variation, and avoid the sample selection issues described by Gray et al. However, they suffer from small sample size, attrition, and their effects are at risk of being driven by disparities between specific SIPP years.

Sample composition, stability, and size are key challenges for the stock samples. While on the left side of the age thresholds I observe all individuals who were observed in the specified age range (48.5-50 or 58.5-60), on the right side I only observe the individuals from the left side who remained in SIPP after age 50 or 60. This means that sample size declines in age, as shown in Appendix Figures A3C and A3D. While this hurts precision, especially at the upper end of the age range, this will not cause bias if the probability of an individual attriting from the sample is not associated with their potential outcomes. Unfortunately, I cannot impose balance on the sample by restricting to individuals observed on both sides of the cutoff without making the sample too small to be viable. The stock samples cover an age range of 3 years or 36 months. Meanwhile, the 2001 SIPP panel only observed individuals for 36 months. This means that the only individuals from the 2001 panel who are present at all ages of the stock samples are those who were exactly 48.5 (or 58.5) at the start of the first wave of the panel. Meanwhile, people who were 50 at the start of the first waive can remain in the stock sample until the end but will be missing from the lower end of the sample's age range. This means that the only observations on the higher and lower ends of the age range from the 2001 panel are from a small number of specific birth cohorts and sample months. Though it covered 48 months, and thus its contributions to the stock samples are somewhat more balanced across age, the 2004 panel presents a major challenge to sample stability. Due to funding issues, SIPP was forced to drop approximately half of its participants, and thus they are missing from the last 1.5 years of the panel. The wave structure of SIPP meant that this attrition occurred gradually between May 2006 and September 2006, resulting in many individuals only being observed for a maximum of 32 months and thus cannot appear at all ages of the stock samples. It is unclear what effect this had as it was unclear how they decided who to drop. Finally, the 2008 panel observed individuals for up to 64 months. While the 2008 panel didn't include significantly more individuals than the 2004 panel, this longer time-span means that more of them were observed in the stock samples' age ranges. This means that the 2008 panel's contributions to the stock samples are more stable and come from a larger number of years and cohorts, but also means that the 2008 panel dominates the stock samples, especially on the extreme high end of the samples' age range.

I also construct "combined" samples in an attempt to generate alternate estimates which do not suffer from the small size and compositional issues of the stock samples. These samples contain all person-months in a 6 year age range centered on each sample's relevant age cutoff which otherwise met criteria for inclusion in the samples. For the ABAWD samples this would be observations with ages greater than or equal to 45 and less than 55. As individuals may join these samples after passing their age thresholds, it no longer makes sense to drop individuals based specifically on their characteristics from before 50. The combined samples have the benefits of providing a large sample size and allowing for a smoother distribution of observations across age. Unlike the stock samples, ages further from the cutoff will not be based on a dwindling number of individuals the further we get from the age cutoffs. Additionally, allowing people to enter the sample at any age reduces the risk of sample selection described in the previous paragraph. A major drawback of the combined samples is that, in order to avoid the post-treatment sample selection issues described previously, they instead are constructed in such a manner as to fall victim to substantial sample dilution. While the details of this are described in more detail below, they are designed around the principle of not dropping individuals even if they are unlikely to be subject to a work requirement. This provides a larger sample with a relatively stable number of observations across age. Thus I have a stock sample for the general req, a stock sample for the ABAWD req, a combined sample for the general req, and a combined sample for the ABAWD req.

SNAP is a program intended for low income families, and thus its rules are primarily relevant to those with incomes low enough to qualify or incomes which might place them at the margin of participation. All of my samples are restricted based on income, only including observations from individuals for whom more than 10% of their observed months were spent with household, family, or subfamily income below 250% of the federal poverty line. SIPP's income variables may not always be precise, as evidenced by the fact that there were 210 individuals in the combined General sample who reported receiving SNAP at some point but who were never under 130% of the poverty line, in spite of this being a requirement for program eligibility. For this reason I also include observations from individuals who ever lived in a family where at least one person was receiving SNAP. Unless stated otherwise, characteristics are based on all months in which a person was observed in the SIPP data, not just the months which were included in a given sample. In total I have 4 samples, 2 for each Work requirement: one "stock" sample and one "combined"/"unfiltered" sample.

Individuals are exempt from a work requirement if they or their family meet certain criteria, and thus the work requirements are irrelevant to them. In order to avoid dilution, past papers, which primarily observed individuals only once, excluded individuals who were exempt. However, the decision of who to drop becomes more difficult in the context of a panel dataset because an individual may be exempt, or not exempt, in different months. While those always exempt can be dropped, and those never exempt can be kept, it is less obvious how to treat individuals who spent 3 out of 12 months exempt and thus the choice is somewhat arbitrary. For this reason, I exclude individuals from the sample based on having a high proportion of exemptions, which is defined slightly differently in different samples. While I cannot directly observe exemption from a work requirement, I approximate whether they would be subject to a given work requirement based on their observed household characteristics. Individuals are exempt from the General Requirement if they are caring for a child under 6 or a disabled family member. I identify a person-month as satisfying this exemption if they are the only adult in a household with a child under 6, or if they are 1) not themself disabled 2) currently lived with an individuals who was under 6 or disabled, and 3) reported that their primary reason for not working was "Taking care of children/other persons" (ERSNOWRK=6). I also consider someone exempt from the General Requirement if they are disabled. I consider individuals to be exempt from the ABAWD Requirement in a given month if they lived with a child in their family, or if they were exempt from the General Requirement. The second set of exemptions encompasses the first because they overlap in terms of disability and because, as past papers have largely failed to recognize (except for Han 2022), exemption from the General Work Requirement is a sufficient condition for exemption from the ABAWD requirement. An individual is defined as being disabled in a given month if they were either "Unable to work because of chronic health condition or disablity" (sic) (ERSNOWRK=3) or "Had work-preventing physical/mental/health condition" (EDISPREV=1). An individual is defined as living with a child in their family if “Total number of children under 18 in family” was greater than 0. In the context of sample construction, I do not consider someone as exempt if they were receiving UI, though this would exempt someone from both work requirements. This is because receiving UI is a relatively transitory condition compared to disability or the composition of one’s household. Additionally, UI may be endogenous to treatment and outcomes and I wished to avoid sample selection.

For stock samples involving the General Requirement, I drop all observations from an individual if they were exempt for $\geq 50\%$ of their observed months before the age of 60. Thus, I only include individuals who spent a majority of their pretreatment observations subject to the General Work Requirement. For stock samples involving the ABAWD requirement, I drop all observations from an individual if they were exempt for $\geq 50\%$ of their observed months before age 50. Exclusions based on these criteria are based on all observed months before the age of 50 (or 60) not just the months included in the sample. I drop individuals based on their characteristics before the age thresholds because the estimands (described in the empirical section) are the effects on those who were subject to the work requirements prior to hitting the age cutoff. Thus, I only consider pretreatment exemptions because, as described above, some proportion of the people who were subject to a given requirement would have become exempt from that requirement after turning 50, thus reducing the true treatment effect.

Given that many individuals in the combined samples were not always observed before the relevant age cutoff, it no longer makes sense to drop individuals based on exemptions before the cutoff. However, as discussed in this section, selecting individuals based on exemptions on either side of the cutoff is likely to lead to selection bias. Instead, I include all individuals and rely on controls for exemption status to account for this source of endogeneity. Thus, individuals are not more or less likely to be observed before and after aging out of a work requirement. A drawback of this approach is that it invites sample dilution as the combined samples will include a substantial number of individuals who were never subject to any work requirement.

The largest limitation of my samples is that I cannot directly observe or account for ABAWD Work Requirement waivers or 15% exemptions. There is unfortunately no readily usable dataset which tracks SNAP geographic waivers, and even if there were, SIPP suppresses the geographic variables necessary to match individuals to waived areas. Meanwhile, 15% exceptions are often applied at the individual level and are not recorded in SIPP. Additionally, the criteria which states use to apply these exemptions has largely not been documented, and even information on the extent of waiver usage is not easily accessible. These waivers and exemptions will mean that a large proportion of my ABAWD samples will not be subject to the ABAWD requirement at any given time. While it is hard to give an exact estimate, it is likely that approximately 50% of the observations under 50 in my ABAWD samples were not subject to the ABAWD requirement (Cuffey et al. 2022). Additionally, even if I were able to observe waivers and exemptions, restricting my sample to those who were under the requirement would likely make it too small to be viable. While there are several ways to conceptualize this, the practical implication is that estimates based on my ABAWD samples will be severely biased towards 0. If the treatment is the cessation of the ABAWD requirement, then a large proportion of the samples were treated prior to 50, rendering the estimates merely intent to treat.

Descriptive statistics before and after the age threshold for each sample can be found in Table 1. Further descriptive figures can be found in Appendix Section 5.

Table 1: Summary Statistics

	Combined Gen.		Combined ABAWD		Stock Gen.		Stock ABAWD	
	Under 60 mean	Over 60 mean	Under 50 mean	Over 50 mean	Under 60 mean	Over 60 mean	Under 50 mean	Over mea
SNAP Participation	0.089	0.086	0.075	0.073	0.028	0.035	0.027	0.03
Employed	0.571	0.480	0.732	0.695	0.737	0.677	0.860	0.84
Disabled	0.252	0.277	0.158	0.179	0.037	0.072	0.020	0.03
Family Exemption	0.010	0.010	0.019	0.017	0.003	0.004	0.006	0.00
Lived With Children	0.166	0.126	0.441	0.338	0.149	0.127	0.431	0.38
UI Benefits	0.032	0.025	0.029	0.028	0.038	0.037	0.035	0.03
Male	0.454	0.441	0.464	0.463	0.466	0.456	0.478	0.48
White Alone	0.775	0.788	0.763	0.761	0.799	0.804	0.774	0.76
HS Degree or Less	0.489	0.513	0.501	0.491	0.442	0.435	0.468	0.47
Income as % of FPL	2.704	2.658	2.704	2.787	3.025	2.975	2.976	3.01
Medicaid	0.146	0.144	0.133	0.134	0.045	0.051	0.056	0.06
Observations	216,815	204,885	141,470	129,535	78,138	58,797	57,947	40,72

6 Empirical Strategy

The ultimate goal is to estimate the difference in outcomes in older adults between 2 states of the world: the world with work requirements and the world without work requirements. The specific estimand of interest is the effects that the removal of the ABAWD and General Work Requirements have on program participation and employment among those who, at 49 and 59 respectively, would be subject to those work requirements and for whom SNAP is relevant. Specifically, I am interested in those who, at the ages of 49 or 59, are either subject to those requirements because they are on the program or who are not on the program but would be on it if it weren't for the existence of the work requirements. This would exclude effects on employment and reduced attrition for those who were exempt from the requirement prior to 50 or 60, but who would have lost exemptions afterwards (ex: disabled at 49 but became non-disabled at 50). For each work requirement and outcome, the effects of interest can be decomposed into 2 distinct effects on 2 distinct groups: those already on SNAP before the removal of the work requirement (incumbents), and those not already on SNAP (non-incumbents).

As the treatment of interest, the removal of work requirements, occurs due to an arbitrary policy cutoff at a specific age. Thus an RD design recommends itself. Based on Lee and Lemieux, RD relies on the assumption that the expected values of the potential outcomes are continuous in the running variable at the cutoff. "In essence, this continuity condition enables us to use the average outcome of those right below the cutoff (who are denied the treatment) as a valid counterfactual for those right above the cutoff (who received the treatment)." The assumption of no fine manipulation of the running variable, which is implied by this, is likely sustained as it is impossible for one to change their age and it is unlikely that anyone would go to the length of successfully falsifying their age in order to be free of a work requirement a few months early (and then also lie about their age on an unrelated survey). However, additional assumptions are required in the context of panel data. Now, we must assume that the average outcome while the work requirement is still in place (before 50 or 60) is a valid counterfactual for average outcome at a later age, after an individual has experienced the first outcome and now has had the work requirement removed, conditioning on age itself. This requires that having already experienced the pretreatment outcome does not, in and of itself, affect the potential post treatment outcome. Additionally, foreknowledge of the impending treatment must not affect the pretreatment outcome, or indeed the post treatment outcomes. We require 2 additional identifying assumptions: the outcomes on either side of the cutoff may not affect each other, the foreknowledge of the treatment does not affect potential outcomes on either side of the cutoff. Additionally, potential outcomes may not be correlated with sample composition/attrition and regressions involving fixed effects require the assumption of strict exogeneity. Though I shall discuss potential violations of these assumptions, it is important to discuss what these assumptions will produce. As discussed by Lee and Lemieux, using age as the running variable when there is inevitable treatment precludes interpreting an RD as analogous to a randomized experiment which produces the ATE. Nearly all individuals age across the cutoff, and thus all are in the "treatment group". Thus we can view these RD models as estimating the TOT, looking at the effect of the removal of the work requirement given that someone is going to experience the presence and then absence of the work requirement. However, as mentioned previously, regressions involving the ABAWD requirement will, at best, estimate the ITT.

The primary identifying assumption for the RD models is that, conditional on exemptions and fixed effects, the potential outcomes are continuous in age across the doughnut. Stated slightly less accurately, this can be seen as an assumption that an individual 3 months before 50 (or 60) will be the same as themself 3 months after turning 50 (or 60) (in terms of their potential outcomes), after controlling for exemptions and age itself.

There are several major empirical challenges in estimating the effect of aging out of a work requirement. Program participation and employment are likely to change as people age, independent of work requirements. There are also two key issues, discussed in the theory section, which are likely to bias a sharp RD towards 0: anticipation of aging out of work requirements and dynamic treatment effects. The former of these represents a violation of the assumption of no response to foreknowledge of treatment. The latter of these is compounded by an inability to distinguish between the effect of treatment and the effects of aging. It is likely that both the anticipatory effects and dynamic treatment effects will be most heavily concentrated in months surrounding the age cutoff. Thus, to ameliorate these issues, I will use a doughnut design which excludes observations in the 3 months preceding, and 3 month following, an individual's relevant birthday. Thus, I will exclude observations aged 597-602 months from regressions involving the ABAWD Work Requirement, and those aged 717-722 months for the General Requirement.

There is a limit to how far in advanced people can apply for SNAP or reduce their employment in anticipation of aging out of a work requirement. On the most basic level, a work requirement is a requirement which must be met in order to receive benefits from, and remain on, SNAP, and thus they limit anticipatory changes in participation and employment. If people apply far in advance of turning 50 or 60 while not meeting the requirement then they will be kicked off before aging out. This will cut down on the aggregate quantity of anticipatory applications through program exit and through deterring application. It is possible that individuals may preemptively apply with the intention of initially meeting the work requirement and then stopping their compliance upon aging out of it (or shortly before). However, this would require inconsistent preferences. Regarding the period in which they briefly meet the requirement, if they would prefer being on the program while meeting the requirement (and were able to do so) over not being on the program while not meeting the requirement then they would have already been on the program in the first place. Analogous arguments apply to limiting the anticipatory employment response. Thus, theoretically, the nature of having a work requirement limits the extent to which anticipatory responses are feasible. Additionally, several specific program features which contribute to this. In the absence of waivers and exemptions, an individual not meeting the ABAWD requirement may only receive SNAP for a maximum of 3 months. Additionally, for working heads of household approaching 50 years old and who are subject to the General Requirement, the prohibition against voluntary quits is retroactive to 60 days before they apply for SNAP. Thus there are reasons to think that anticipatory responses to the ABAWD requirement will be significantly concentrated in, though not necessarily restricted to, a window of the 3 months before turning 50 and therefore excluding these 3 months will reduce this bias. On a less formal note, if the concern is that people will act in anticipation with the knowledge they will soon age out of a work requirement, this requires that they will SOON age out of the work requirement and so this effect is likely concentrated closest to that threshold. This applies to both work requirements. While the choice of 3 months is admittedly more arbitrary in the case of the General Requirement, their exclusion will likely reduce some of the anticipation bias.

Excluding the 3 months following aging out of a work requirement means that an RD estimate will aggregate the effects of the first 3 months of treatment. This means that it will capture both the immediate and dynamic effects which occur in these months. Thus, this will reduce the bias caused by dynamic treatment effects, to the extent that they are concentrated in these 3 months. Some of the dynamic treatment effects may be concentrated in the months following turning 50 or 60. While I can think of no reason why the effects of reduced program attrition, as described in the theory section, would be concentrated in the month following treatment, the effect of program entry may be concentrated in this window. Using the ABAWD requirement as an example, there are several types of people who will drive a participation response via program entry. Some individuals will know that they are now eligible and will apply immediately upon turning 50 and thus their change in participation may occur in the month they turn 50 or the next month. Some of them may live in a place with an application backlog and so it takes 2 months. Others may be broadly aware of the work requirement, but may delay their response. Suppose someone wants SNAP benefits but doesn't remember that they are now eligible a couple of weeks after turning 50, doesn't get around to applying for another week, and doesn't get approved for a few more weeks. For both of these types of people, their participation response may not materialize immediately at the age of 50 but will occur within a few months of turning 50. Meanwhile, someone else may have a $0 < p < 1$ probability of finding out they are eligible and applying in any given month. The full participation response of this group will likely be drawn out beyond the first 3 months. More hypothetically, perhaps p may be slightly higher in the first few months after their birthday due to friends on SNAP being more likely to mention the new change that occurs at 50. If this were the case then their participation response, while still drawn out, would be slightly concentrated at 50. Taken together, while the full dynamic effects of program entry will not occur immediately after turning 50 or 60, removing the 3 months following these birthdays is likely to significantly reduce the bias in program participation caused by dynamic treatment effects. Dynamic effects on employment which are associated with program entry are likely to loosely follow the same time patterns as program entry, and thus will reduce the corresponding bias. However I do expect employment response to be at least somewhat more drawn out as people's preferences for work change as they age. The timing of the employment response among incumbents is less clear. Likely some will immediately reduce employment (group 1), some will find out about the change immediately but may take a couple of months to respond (such as if they are finishing up the current season of seasonal work)(group 2), and some will only find out and respond much later (group 3). The first group will be captured by the RD with or without the doughnut, the second group will only be captured with the doughnut, and the third group will remain part of the bias towards 0. Excluding the 3 month doughnut will reduce dynamic treatment bias in employment among incumbents to the extent that there are individuals in group 2. Finally, the choice of 3 months after treatment is arbitrary but the choice of, and adherence to, a specific arbitrary time-span avoids the tendency to manipulate results.

I have 2 dependent variables of interest: individual participation in food stamps (ER27=1 in SIPP), and individual employment (EPDJBTHN=1). Regressions will be run separately, in separate samples, for each work requirement. Thus, age cutoffs should be interpreted as 50 for the ABAWD requirement and 60 for the General Requirement, and the running variables are normalized with regard to this. I shall do 3 sharp regression discontinuity specifications. The first, based on Cellini, Ferreira, and Rothstein 2010, is a sharp RD estimated using STATA's xtreg and includes individual and month fixed effects.

$$Y_{it} = \alpha + \beta_1 D_{it} + \gamma_1 a_{it} + \gamma_2 a_{it} D_{it} + \theta_i + \lambda_t + \delta_1 K_{it} + e_{it} \tag{1}$$

Here, Y_{it} is employment or SNAP participation for individual i in month t. The running variable a_{it} is age in months relative to the appropriate cutoff (600 or 720 months old), while D_{it} is an indicator that an individual has crossed the cutoff. θ_i and λ_t are person and month fixed effects, respectively. K_{it} is a series of indicators that someone is currently exempt from the relevant work requirement. This includes disability, caring for family members, receiving unemployment insurance benefits, and, in the case of the ABAWD samples, living with a child under 18. Standard errors are clustered on the running variable. Thus, β_1 gives the treatment effect of aging out of a work requirement.

As it is admittedly unusual to include individual fixed effects in an RD, I perform 2 additional sharp RDs identified on between-variation. The first is mostly identical to equation 1 but without the fixed effects.

$$Y_{it} = \alpha + \beta_1 D_{it} + \gamma_1 a_{it} + \gamma_2 a_{it} D_{it} + \rho_i + \delta_1 K_{it} + X_i + e_{it} \tag{2}$$

Standard errors are clustered at the individual level. This regression estimates the effects of ending the work requirements using between-variation. X_i is time invariant individual covariates, and ρ_i represent SIPP panel fixed effects.

Additionally, following some of the recent literature, I will estimate a sharp RD using the methodology recommended by Cattaneo, Idrobo, and Titiunik 2019. This will use linear regression local to MSE optimal bandwidths about the edges of the doughnut with triangular kernel. Estimation is performed using the rdrobust STATA package/command. While this specification is arguably nonparametric, it is equivalent to performing the following regression in a bandwidth on either side of the doughnut, with the effect of interest being the difference between their intercepts.

$$Y_{it} = \alpha + \gamma_1 a_{it} + \delta_1 K_{it} + X_i + e_{it} \tag{3}$$

Standard errors are calculated using the authors' Robust Bias Correction formula and clustered at the individual level.

Concern about anticipatory effects and dynamic treatment effects leads me to adopt a 2WFE specification.

$$Y_{it} = \alpha + \beta_1 D_{it} + \theta_i + \lambda_t + \delta_1 K_{it} + e_{it} \tag{4}$$

While this RD design would estimate the average difference in the locality of the cutoff (or rather the doughnut), under this model I would be estimating the difference in average outcomes between those below and above the treatment age (and outside the doughnut). (or rather an aggregation of 2 way comparisons per Goodman-Bacon). The anticipatory effects in the months just before 50 would still cause bias as they would still be included on the left side of the cutoff, though they would be diluted by averaging them with months further from the cutoff. Any dynamic effect of treatment to raise food stamp participation gradually over time would likely partially raise the mean outcome after treatment, and thus may partially be captured by a 2WFE model. However, as has been well established (ex: Goodman-Bacon 2021) 2WFE does a poor job capturing dynamic treatment effects. Additionally, misspecification resulting from these dynamic treatment effect would lead a 2WFE model to suffer from positive serial correlation, and thus increased risk of type 1 errors.

While the above models provide an estimate of the effect of being above or below an age cutoff, without a comparison group I cannot differentiate between the effects of treatment and the effects of aging. For example, figure 3.B.1 shows that employment declines after the age cutoffs. However, it is unclear whether this is truly the effect of treatment given that it was also declining before the age cutoffs. In an attempt to better identify the average treatment effect, I specify one or more comparison groups for each sample, which I then use to estimate several variants on a standard difference in difference model. For the unfiltered samples, I select individuals who spent less than 30% of their observed months in the sample exempt from the relevant work requirement (low exemption), and individuals who spent at least 70% of their time exempt (high exemption). As we would expect the effects of aging out of work requirements to be concentrated among those who were subject to them, those with low exemptions serve as the treatment group ($T_i = 1$) and those with high exemptions serve as a comparison group ($T_i = 0$). For the stock samples I define low and high exemptions groups based on exemption rates before turning 50 (or 60). Thus, I restrict the stock samples to those who were exempt for less than 30% of their observed month before reaching the age cutoff. I then combine this with a high exemption comparison group, constructed analogously to the applicable stock sample. I then estimate the following models in each sample. The first is a DID model based on between-variation and the second is a DID model based on within-variation.

$$Y_{it} = \alpha + \beta_1 D_{it} + \beta_2 D_{it} T_i + \beta_3 T_i + \rho_i + \delta_1 K_{it} + X_i + e_{it} \tag{5a}$$

$$Y_{it} = \alpha + \beta_1 D_{it} + \beta_2 D_{it} T_i + \theta_i + \lambda_t + \delta_1 K_{it} + e_{it} \tag{5b}$$

Where T_i is an indicator that someone is in the treatment group, and thus β_2 is the effect of interest in both equations.

The American Recovery and Reinvestment Act (ARRA) waived the ABAWD Work Requirement nationwide April 2009-Sept 2010 (Cuffey, Harris). Due to the slow economic recovery after The Great Recession, nearly every state and county continued to be waived through 2013 (Harris). Thus we would not expect those turning 50 in these years to experience a discontinuous change in incentives or program rules. Therefore I estimate a version of equation 5 for the ABAWD requirement using the original stock ABAWD sample as the treatment group and an analogously constructed stock sample of observations from April 2009 to November 2013 from the 2008 SIPP panel as a comparison group.

For all of these DID regressions, a key identifying assumption is parallel trends. Additionally, these DID regressions require assumptions, analogous to those of th RD regressions, in order to be unbiased: no spillovers between outcomes before and after treatment, and the foreknowledge of the treatment does not affect potential outcomes on either side of the cutoff, and strict exogeneity in the regressions which include fixed effects.

7 Results

7.1 Regression Discontinuities

Table 2

Covariant Balance Regressions, Coefs. of Interest

	Samples			
Dependent Variables	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
White Alone	0.002	-0.008	0.005	-0.015**
	(0.005)	(0.007)	(0.005)	(0.007)
Male	-0.007	0.011	-0.003	0.008
	(0.006)	(0.009)	(0.006)	(0.008)
Ever HS. or Less	-0.005	0.009	-0.002	0.013
	(0.006)	(0.009)	(0.006)	(0.008)
Ever Military Service	-0.005	-0.002	-0.006	-0.005
	(0.004)	(0.005)	(0.004)	(0.004)
Observations	386,907	248,223	112,029	80,261

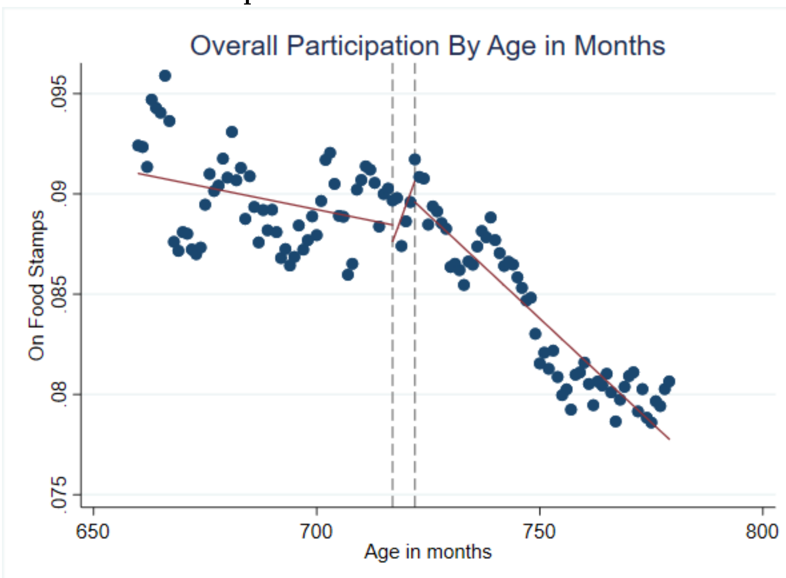
Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

For each sample, McCrary density tests do not find evidence of bunching in the vicinity of the cutoff, or on either side of the doughnut. The exception to this was the sock General Requirement sample, for which the test was weakly significant at p¡0.1 in the vicinity of 60. However, it was not significant about the doughnut, which is the basis for all my RD regressions. When assessing covariate balance in an RD setting, it is necessary that the covariates be determined prior to treatment. Due to the panel nature of my data, I am limited to covariates which were determined prior to the period of my samples, and which are thus fairly constant within individuals over this period. For this reason it would be inappropriate to use the within-variation of equation 1, and instead I regress being white alone, sex, ever having had a high school diploma or less, and ever having served in the US military on the regression discontinuity equation 2 in order to test for balance. The results, which can be found in Table 2 largely do not find discontinuous changes in these characteristics at 50 and 60. The exception is one regression which found that the proportion of the stock ABAWD sample which is white decreases significantly at 50.

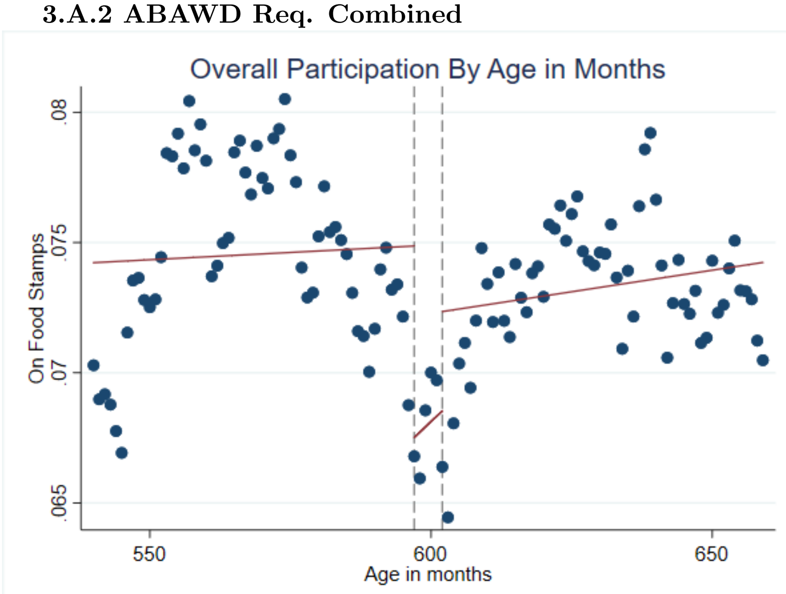
7.2 Figure 3

7.2.1 Figure 3 A

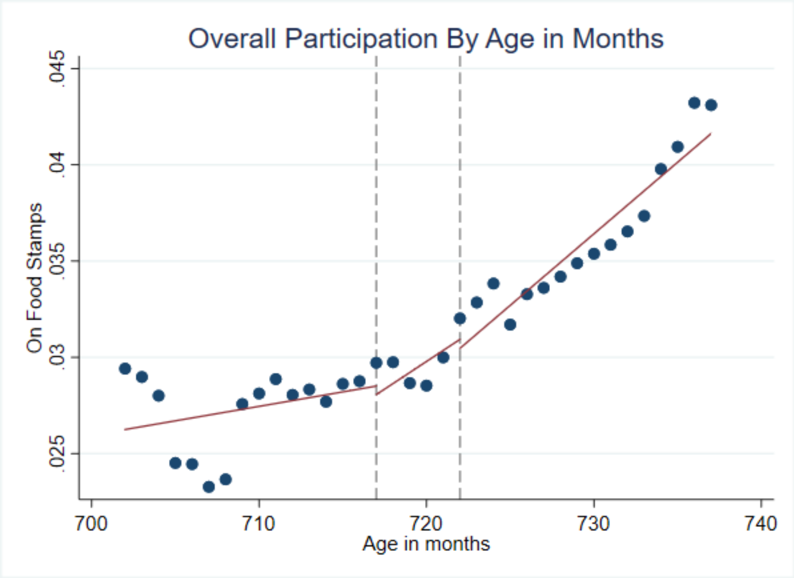
3.A.1 General Req. Combined



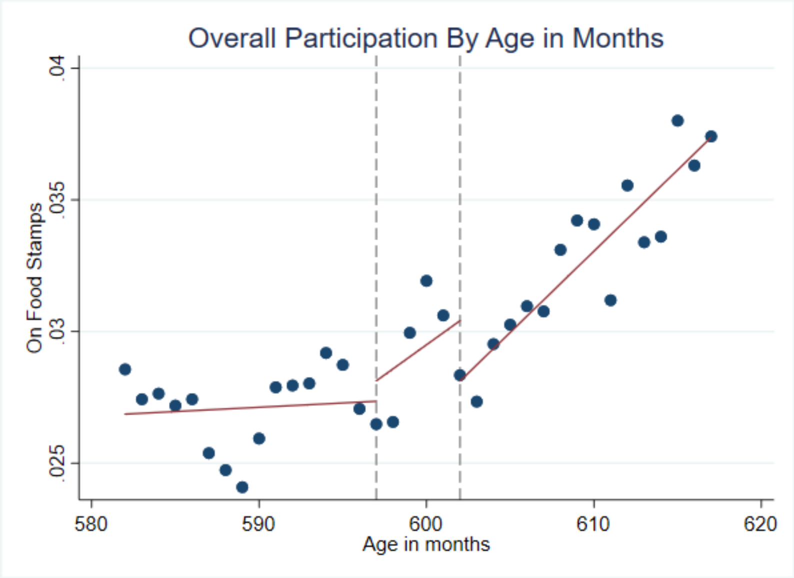
3.A.2 ABAWD Req. Combined



3.A.3 General Req. Stock

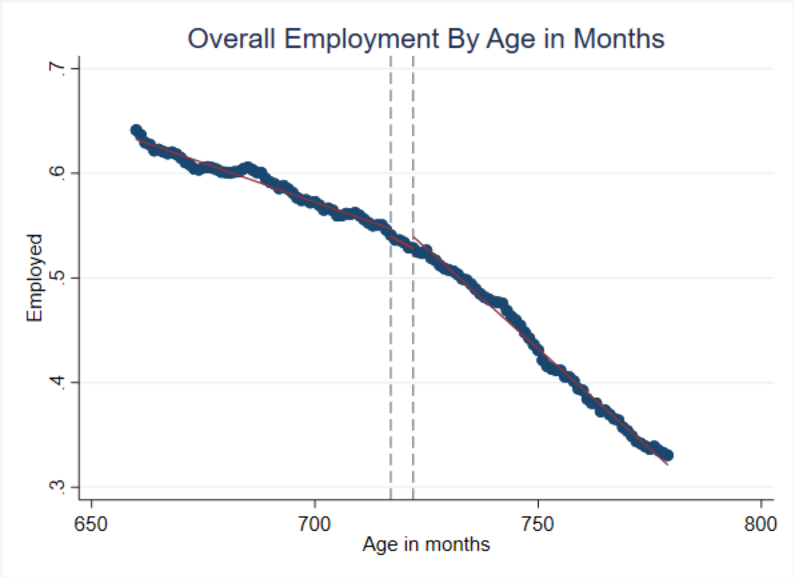


3.A.4 ABAWD Req. Stock

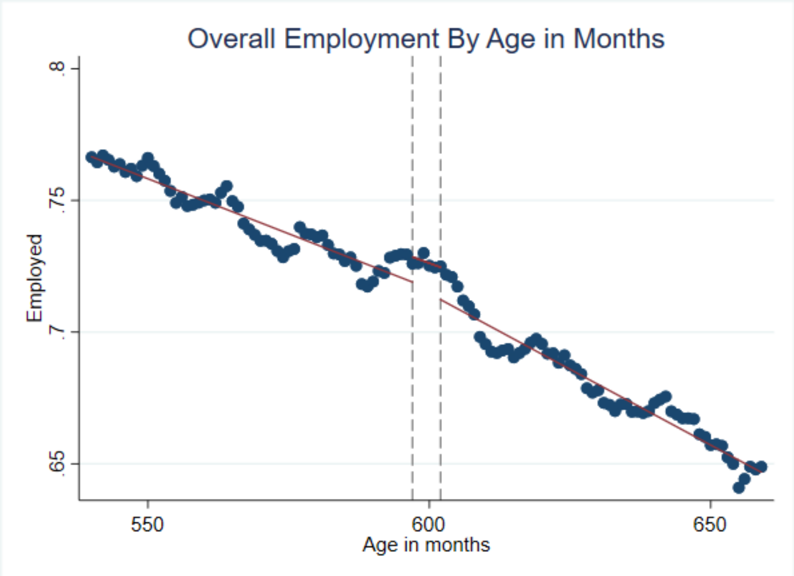


7.2.2 Figure 3 B

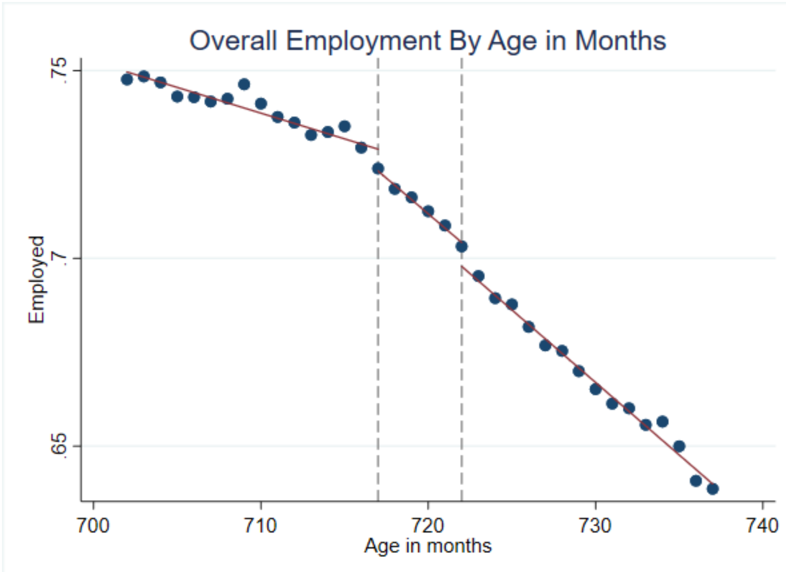
3.B.1 General Req. Combined



3.B.2 ABAWD Req. Combined



3.B.3 General Req. Stock



3.B.4 ABAWD Req. Stock

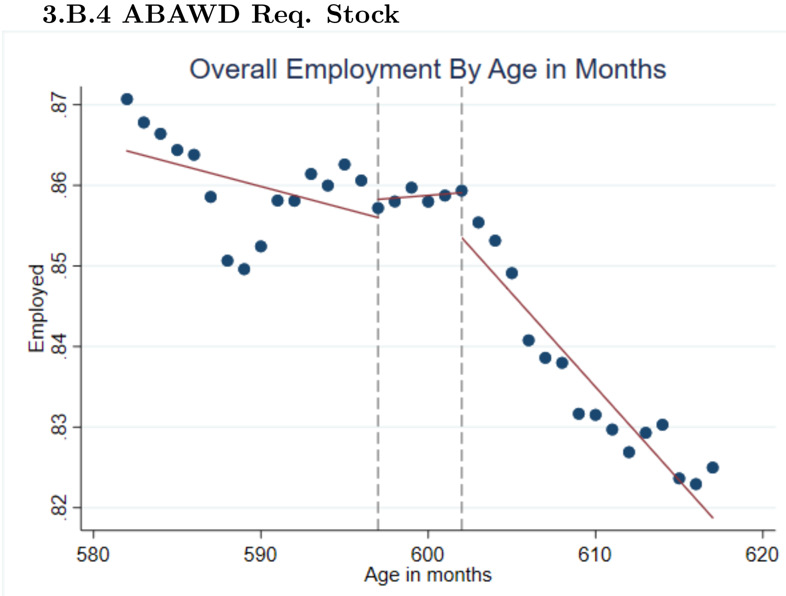


Figure 3 shows binscatters of SNAP participation and employment with respect to the age cutoff for each sample (600 months for the ABAWD Requirement and 720 months for the General Requirement). Observations are binned based on discrete age in months. For the purpose of observing trends in outcomes over a longer time-span, the combined samples' graphs depict observations from a full 5 years on either side of their cutoffs, rather than the 3 years used in estimation. To a significant degree they do not tell a clear story. The General Requirement combined sample (Figure 3.A.1) does seem to indicate an increase in participation by the end of the doughnut, followed by a steep decline, though the high degree of variation both before and after treatment makes it hard to confidently attribute this to the removal of the work requirement. Participation in the combined ABAWD sample exhibits a puzzling sine wave like pattern, with a trough in the vicinity of the age cutoff. It is unclear why participation would decline by 15-20% in the years leading up to 50 before recovering. For both stock samples, participation starts out low but has an upwards trend following the age cutoffs. The General stock sample appears to show a modest increase in participation on the right side of the doughnut, while the ABAWD stock sample has a sudden increase in participation at the age of 50 which disappears after 3 months. Both graphs appear to show a short-lived dip in participation a little over a year before treatment. While none of the participation graphs in figure 3 indicate an unequivocal and discontinuous treatment effect on the right side of the doughnut, there is similarly no clear indication of an anticipatory pre-treatment increase in participation. Additionally, it is noteworthy that all graphs seem to show a change in the slope at the point of treatment. This could potentially provide support for the prediction that the effect of removing the work requirements will manifest gradually over time, though it would be premature to conclude that this interpretation is correct.

The graphs of employment across age (Fig. 3.B) do not provide any indication of a discontinuity at 50 or 60. All are relatively smooth compared to the participation graphs due to the larger degree of variation in employment as compared to SNAP participation. Each shows a downward trend in employment as people grow older, though the magnitude varies. In the stock samples the slope of this trend appears to grow steeper after treatment. While it is possible that a dynamic treatment effect may contribute to post treatment trends in employment, the magnitude of these trends, the treatment trends, and common sense indicate that other causes are substantially responsible.

Overall, the discontinuity plots in Figure 3 do not provide clear evidence of a treatment effect from removing work requirements. However, I caution that these graphs only provide a partial picture as they depict average levels of outcomes across age, and may not be informative about equation 1, which is likely to capture more of the true effect at the cutoff due to using individuals as counterfactuals for themselves. These graphs do not directly reflect this within individual variation across the age cutoff.

Regression results for equations 1 through 3 are shown in tables 3 and 4. While discontinuities and estimates are based on either side of the doughnut, for the sake of brevity I shall refer to them relative to ages 50 and 60. In line with Figure 3, equations 2 and 3 largely do not find a statistically significant change except for one specification of equation 3 (4.C regression 1) which finds a decrease in employment of 1.6 percentage points at 60 (doughnut) (p¡.05) in the combined sample. Equation 1 shows slightly more action for the General Requirement samples, with one specification (3.A regression 1) finding that SNAP participation increases by 0.5 percentage points in the combined sample, or about 6% relative to just before 60 (4.A regression 3), while employment decreased by 1 percentage point after 60 in the stock sample.

This provides some tentative evidence that aging out of the General Requirement, or turning 60, slightly decreases employment and increases SNAP participation. However, the fact that these estimates do not appear to be robust across multiple specifications limits taking them at face value. Nevertheless, there are several caveats to this. Equation 1 uses a different source of variation than equations 2 and 3, and so we would not necessarily expect them to produce similar results. Similarly, the combined and stock samples include very different populations and so their results. It is worth noting that the coefficients from equations 2 and 3 appear to be less precisely estimated than those for 1, resulting in statistical insignificance even when they are similar or larger in magnitude. While this is unlikely to have made a difference in most cases, for the stock General Req. sample the equation 3 effect on employment is larger than the statistically significant equation 1 effect on employment. There are several potential factors contributing to these differences in precision, including: the inclusion of fixed effects, differences in clustering due to an inability to account for all potential sources of autocorrelation in any one equation, and equation 3 having a smaller effective sample and different estimation method.

7.3 Table 3

Table 3.A

FE Regression Discontinuities (SNAP Participation Equation 1)

VARIABLES	(1) UF. Gen.	(2) UF. ABAWD	(3) UR. Gen.	(4) UR. ABAWD
Over Age Threshold	0.005***	-0.001	-0.001	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Disabled	0.030***	0.046***	0.040***	0.035***
	(0.002)	(0.003)	(0.003)	(0.006)
Family Exemption	0.026***	-0.003	0.068***	-0.003
	(0.005)	(0.004)	(0.009)	(0.008)
Children		0.020***		0.012***
		(0.002)		(0.003)
UI Benefits	0.007***	0.005**	0.008***	0.011***
	(0.002)	(0.003)	(0.002)	(0.004)
Constant	0.087***	0.051***	0.039***	0.012*
	(0.004)	(0.004)	(0.007)	(0.006)
Observations	386,907	248,223	112,029	80,261
R-squared	0.008	0.005	0.011	0.004
Number of Individuals	17,585	13,077	6,654	5,258
Individual FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 3.B

Non-FE Regression Discontinuities (SNAP Participation Equation 2)

	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
Over Age Threshold	0.000	0.002	-0.002	-0.002
	(0.004)	(0.005)	(0.003)	(0.004)
Disabled	0.209***	0.257***	0.114***	0.106***
	(0.006)	(0.009)	(0.013)	(0.019)
Family Exemption	0.095***	0.075***	0.080**	0.062
	(0.021)	(0.017)	(0.034)	(0.041)
Children		0.015***		0.020***
		(0.004)		(0.004)
UI Benefits	0.022***	0.020***	0.028***	0.031***
	(0.007)	(0.007)	(0.010)	(0.011)
Constant	0.059***	0.042***	0.028***	0.033***
	(0.007)	(0.007)	(0.007)	(0.007)
Observations	386,907	248,223	112,029	80,261
R-squared	0.137	0.161	0.042	0.031

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 3.C

Cattaneo et al. Regression Discontinuities (SNAP Participation Equation 3)

	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
RD_Estimate	-0.001	-0.004	0.000	-0.001
	(0.005)	(0.006)	(0.004)	(0.004)
Observations	386,907	248,223	112,029	80,261

Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

7.4 Table 4

Table 4.A

FE Regression Discontinuities (Employment Equation 1)

VARIABLES	(1) UF. Gen.	(2) UF. ABAWD	(3) UR. Gen.	(4) UR. ABAWD
Over Age Threshold	0.000	-0.003	-0.010***	0.002
	(0.002)	(0.003)	(0.002)	(0.003)
Disabled	-0.318***	-0.457***	-0.376***	-0.475***
	(0.007)	(0.006)	(0.007)	(0.010)
Family Exemption	-0.277***	-0.352***	-0.380***	-0.448***
	(0.008)	(0.008)	(0.020)	(0.019)
Children		-0.001		-0.013**
		(0.003)		(0.005)
Constant	0.630***	0.791***	0.770***	0.865***
	(0.006)	(0.008)	(0.012)	(0.010)
Observations	386,902	248,223	112,028	80,261
R-squared	0.107	0.116	0.102	0.097
Number of Individuals	17,585	13,077	6,654	5,258
Individual FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 4.B

Non-FE Regression Discontinuities (Employment Equation 2)

	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
Over Age Threshold	0.010	-0.010	-0.005	0.006
	(0.006)	(0.006)	(0.008)	(0.008)
Disabled	-0.680***	-0.815***	-0.636***	-0.694***
	(0.004)	(0.004)	(0.010)	(0.018)
Family Exemption	-0.583***	-0.616***	-0.689***	-0.706***
	(0.018)	(0.024)	(0.018)	(0.041)
Children		0.017***		0.009
		(0.005)		(0.008)
Constant	0.706***	0.824***	0.738***	0.822***
	(0.011)	(0.009)	(0.018)	(0.014)
Observations	386,902	248,223	112,028	80,261
R-squared	0.395	0.506	0.122	0.150

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 4.C

Cattaneo et al. Regression Discontinuities (Employment Equation 3)				
	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
RD_Estimate	-0.016**	-0.008	-0.014	0.001
	(0.007)	(0.008)	(0.010)	(0.009)
Observations	386,902	248,223	112,028	80,261

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 5.A

2-Way FE (SNAP Participation Equation 4)				
	(1)	(2)	(3)	(4)
VARIABLES	UF. Gen.	UF. ABAWD	UR. Gen.	UR. ABAWD
Over Age Threshold	0.005	-0.001	-0.001	-0.001
	(0.003)	(0.004)	(0.003)	(0.004)
Disabled	0.030***	0.046***	0.040***	0.035**
	(0.005)	(0.008)	(0.010)	(0.015)
Family Exemption	0.026**	-0.003	0.068***	-0.003
	(0.012)	(0.012)	(0.024)	(0.028)
Children		0.020***		0.012
		(0.005)		(0.008)
UI Benefits	0.007	0.005	0.008	0.011
	(0.006)	(0.004)	(0.007)	(0.007)
Constant	0.086***	0.031***	0.039***	0.007
	(0.005)	(0.006)	(0.009)	(0.009)
Observations	386,907	248,223	112,029	80,261
R-squared	0.008	0.005	0.011	0.004
Number of Individuals	17,585	13,077	6,654	5,258
Individual FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 5.B

2-Way FE (Employment Equation 4)

VARIABLES	(1) UF. Gen.	(2) UF. ABAWD	(3) UR. Gen.	(4) UR. ABAWD
Over Age Threshold	0.001	-0.003	-0.011	0.001
	(0.005)	(0.006)	(0.007)	(0.008)
Disabled	-0.318***	-0.457***	-0.376***	-0.475***
	(0.008)	(0.013)	(0.018)	(0.026)
Family Exemption	-0.277***	-0.352***	-0.380***	-0.448***
	(0.017)	(0.022)	(0.047)	(0.047)
Children		-0.001		-0.013
		(0.007)		(0.012)
Constant	0.618***	0.817***	0.776***	0.909***
	(0.009)	(0.011)	(0.018)	(0.025)
Observations	386,902	248,223	112,028	80,261
R-squared	0.105	0.116	0.102	0.096
Number of Individuals	17,585	13,077	6,654	5,258
Individual FE	YES	YES	YES	YES
Time FE	YES	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

7.5 Difference-In-Difference

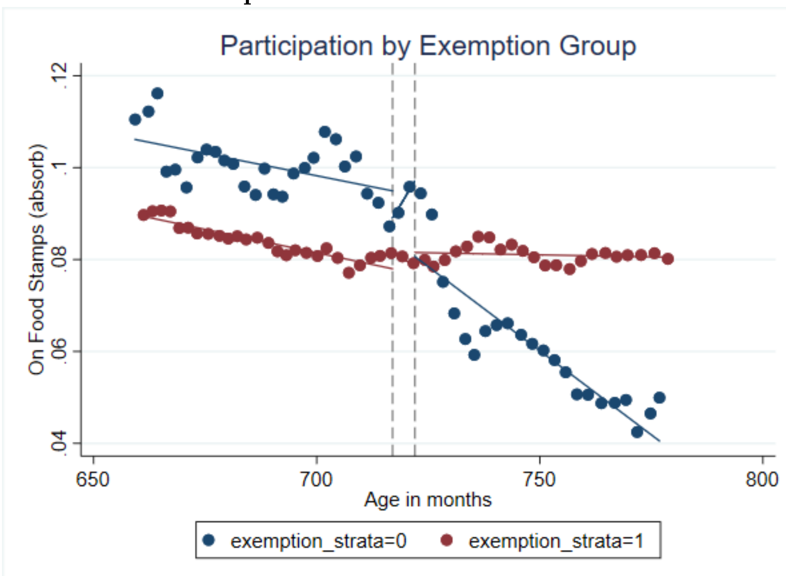
The 2WFE regressions, shown in Table 5, do not indicate a significant effect from removing either work requirement.

Figure 4 shows participation and employment pretends for high and low exemption groups, with outcomes demeaned/residualized by group mean for the sake of visual clarity. As with figure 3, observations are binned based on discrete age in months and the combined samples' graphs depict observations from a full 5 years on either side of their cutoffs, rather than the 3 years used in estimation. Some graphs show clear visual evidence of parallel pretrends, while others do not. The SNAP participation pre-trends appear to be parallel between high and low exemption groups for the combined General Requirement sample, and between 2008 panel and the other panels in the alternate stock sample of ABAWDs. While participation trends are not entirely parallel between exemption groups in the combined ABAWD sample, they are mostly flat for both groups in the years leading up to 50 (these are the years used in estimation). Neither group has a clear trend in the ABAWD stock sample, and thus they are both more or less parallel. For the General stock sample, the high exemption group is not parallel to the low exemption treatment group, though it appears to consist of several parallel segments with discontinuous jumps in between. Given this, the comparison group provides the most plausible counterfactual of SNAP participation for the combined General sample, and the least plausible for the stock General sample.

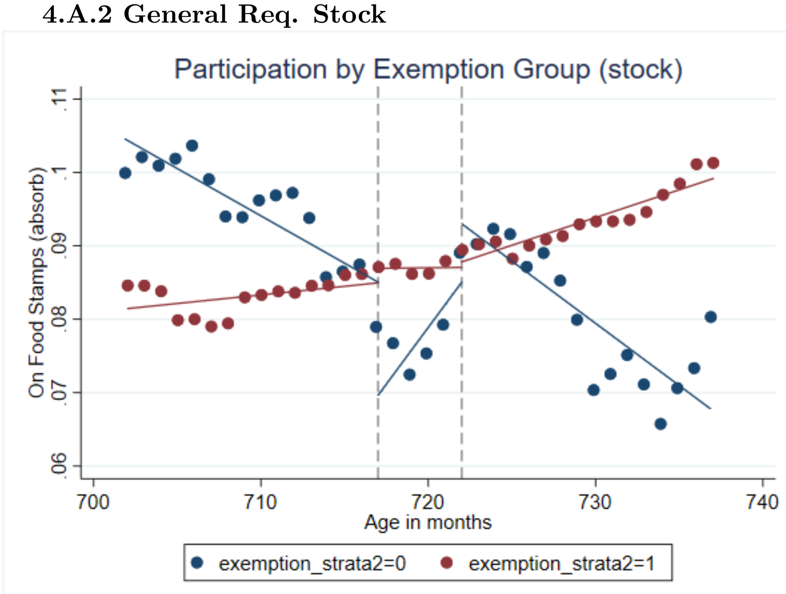
7.6 Figure 4

7.6.1 Figure 4 A

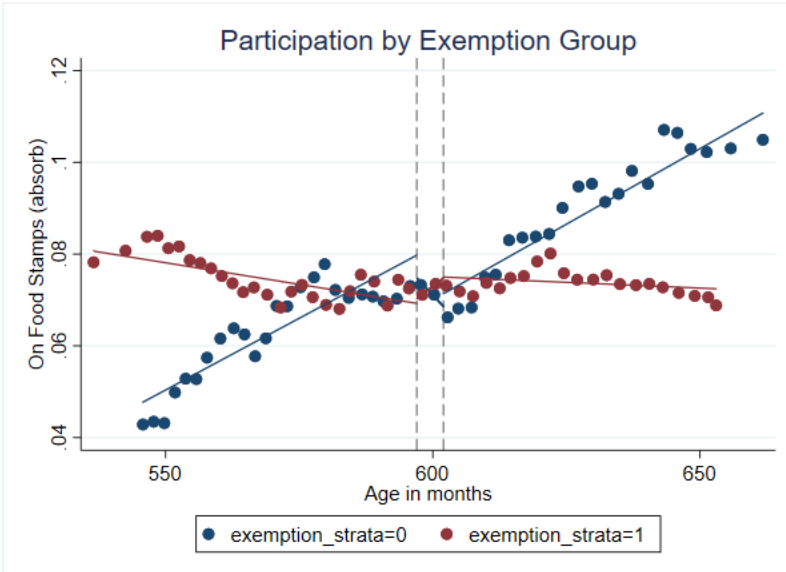
4.A.1 General Req. Combined



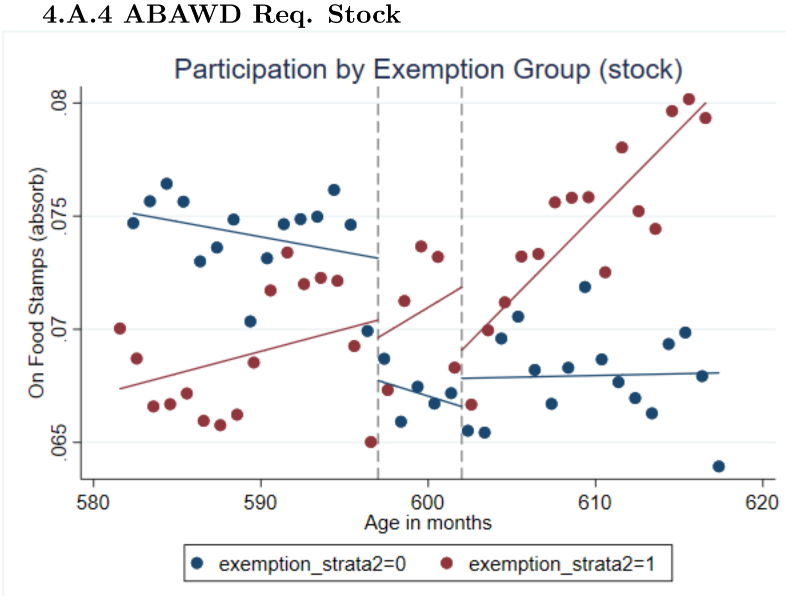
4.A.2 General Req. Stock



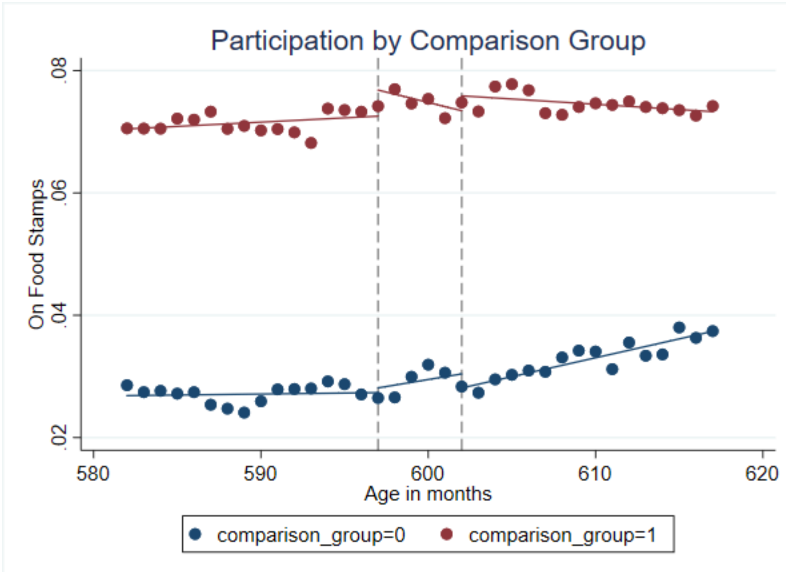
4.A.3 ABAWD Req. Combined



4.A.4 ABAWD Req. Stock

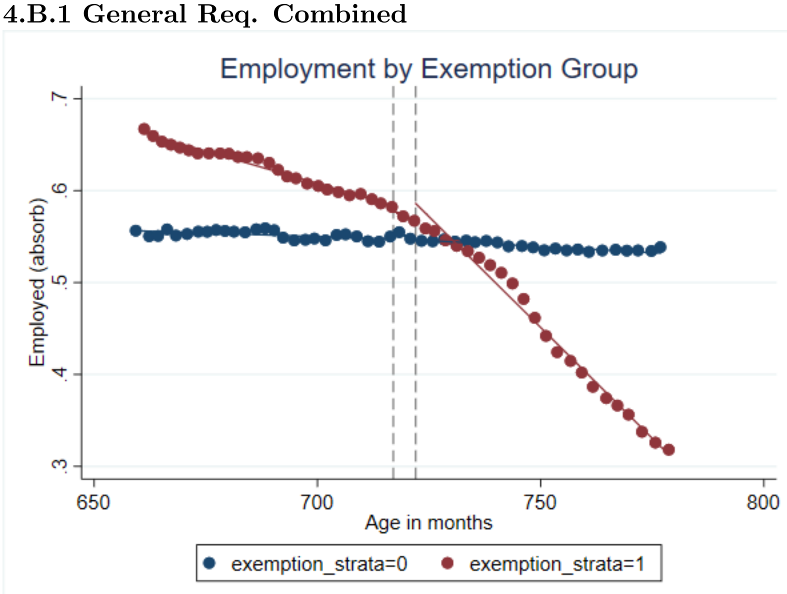


4.A.5 ABAWD Req. Stock Comparison

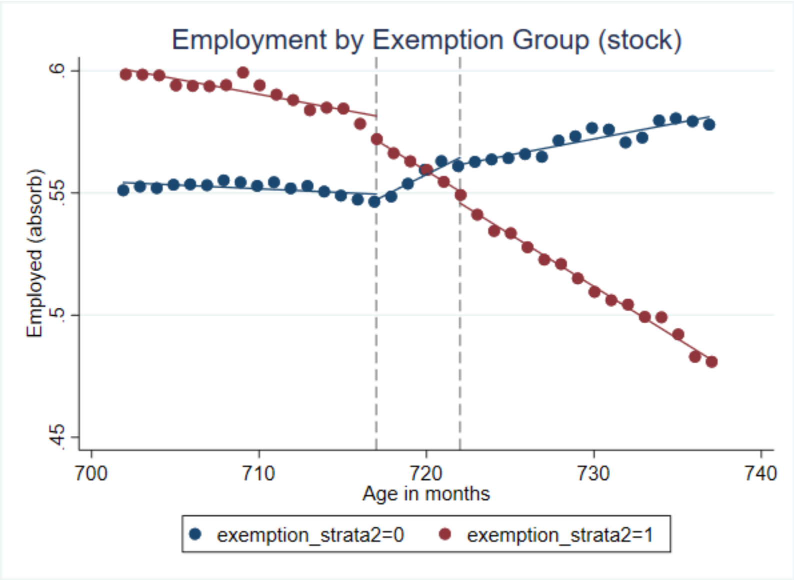


7.6.2 Figure 4 B

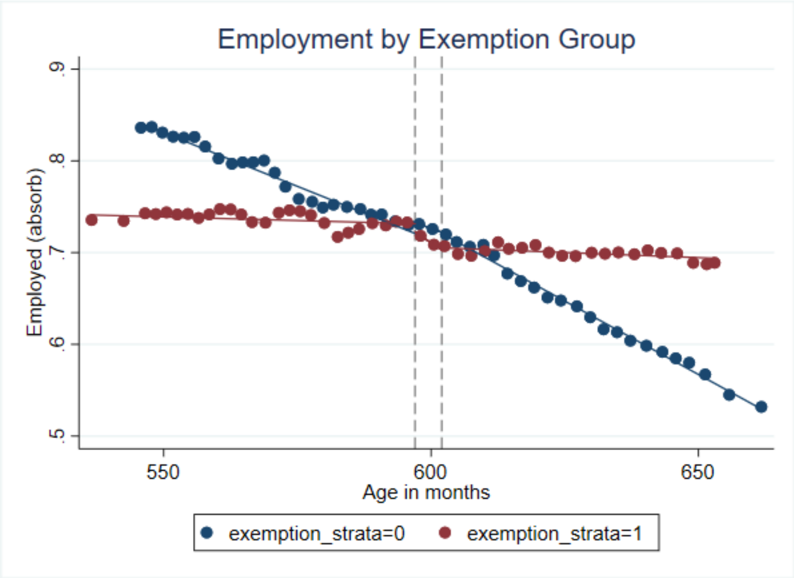
4.B.1 General Req. Combined



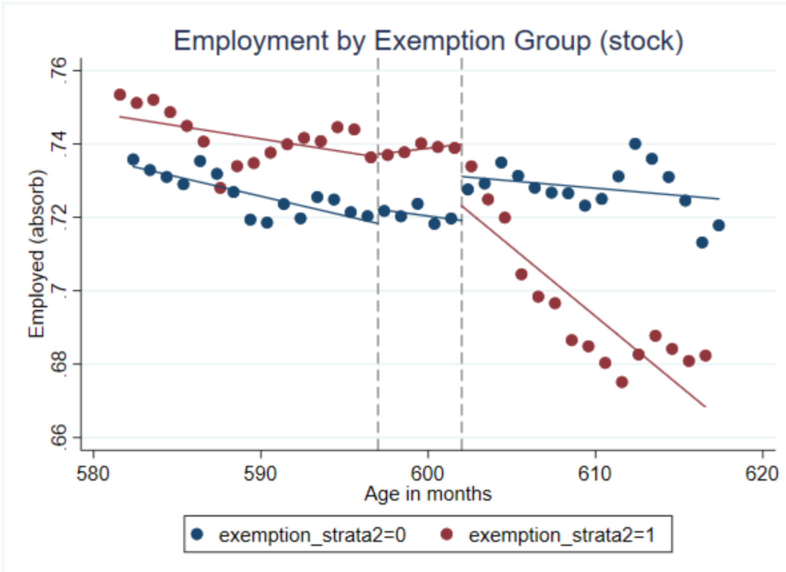
4.B.2 General Req. Stock



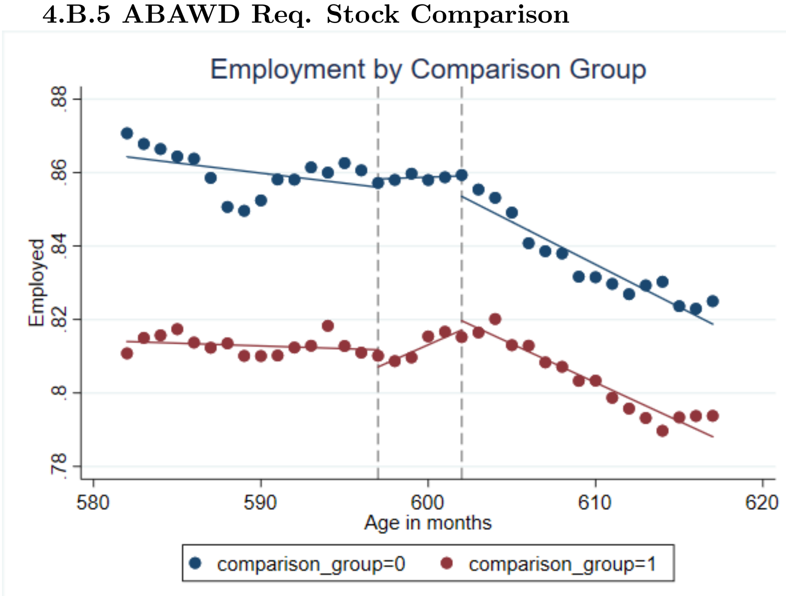
4.B.3 ABAWD Req. Combined



4.B.4 ABAWD Req. Stock



4.B.5 ABAWD Req. Stock Comparison



Here the red group is the treatment group and the blue group is the comparison group.

In the graphs representing employment over age, the pretrends appear to be parallel for both treatment and comparison groups in all stock samples. Meanwhile, they are divergent in the combined samples, though in the case of the ABAWD sample they are both fairly flat in the years immediately preceding 50.

For each sample and regression equation I also formally tested for different pretends by running a specification which included age-in-months dummies, and age-in-months dummies interacted with being in the treatment group (low exemption or before 2008). I then tested the interaction terms for joint significance. For all regressions of SNAP participation, including those based on within-variation, the interaction terms were not jointly significant, meaning that I did not find evidence that the effect of age differed significantly in the pre period between the treatment and comparison groups. For both between and within specifications, the interaction terms were jointly significant for the combined General sample, though only weakly (p¡0.1). This, combined with Figure 4.B.1, leads me to disregard that sample's employment DID regressions.

The results of the DID models are shown in Tables 6 and 7. The removal of the ABAWD requirement is only associated with a changes in participation in one regression, the between-variation regression for the combined ABAWD sample. As the coefficient of interest is only weakly significant (p¡0.1), and the high exemptions group was not a particularly strong counterfactual for the low exemption treatment group, this does not provide clear evidence of the ABAWD requirement having an effect. In regressions identified on between variation, the removal of the General Requirement appears to increase program participation, though this is only significant for the combined sample where turning 60 is associated with a 2.8 percentage point increase in SNAP participation, or about 122% relative to pretreatment levels. However, this magnitude seems improbable and, based on Figure 4.A.1, this effect appears to be driven by a decrease in SNAP participation among the comparison group after 60, while participation in the treatment group remains fairly constant. It is possible that, in the absence of treatment, the treatment group would have experienced a similar decline and that they only experienced constant participation through the countervailing effect of the treatment to increase participation. However, this conclusion would require strong evidence and further falsification as it seems more probable that there is no treatment effect in this sample, and that the comparison group experienced some contemporaneous shock.

Meanwhile, in the within-variation regression, the removal of the General Requirement appears to decrease the proportion of individuals receiving SNAP by 3.4 percentage points (p¡0.01) in the combined sample and 1.9 percentage points (p¡0.05) in the stock sample. These constitute a decrease of 148% and 79%. These percentage changes seem shockingly large, and I discuss them further in my discussion section.

The removal of the General Work Requirement is strongly associated with a decrease in employment (in the stock samples). When using between-variation, employment falls by 3.5 percentage points (p¡0.01) in the stock sample, or about 4.6%. Based on Figure 4.B.2, this occurs as employment falls among those subject to the work requirement after treatment, while, over the same age range, it rises slightly for the comparison group who were overwhelmingly exempt from the requirement. Meanwhile, when using within-variation, employment in the stock sample falls by 5.1 percentage points (p¡0.01), or about 6.7%. Finally, for the ABAWD stock sample, employment decreased by 1.9 percentage points (p¡0.05) in the within regression.

Table 6.A

DID Between-Variation (SNAP Participation Equation 5)

VARIABLES	(1) UF. Gen.	(2) Gen. Stock	(3) UF. ABAWD	(4) ABAWD Stock	(5) ABAWD Comp.
Over Age Threshold	-0.029**	-0.008	-0.011	-0.006	0.003
	(0.011)	(0.011)	(0.007)	(0.007)	(0.005)
Low Ex. x Threshold	0.028**	0.010	0.014*	0.003	
	(0.012)	(0.012)	(0.008)	(0.008)	
Low Exemptions	-0.154***	-0.130***	-0.065***	-0.057***	
	(0.012)	(0.016)	(0.013)	(0.015)	
Disabled	0.102***	0.127***	0.230***	0.225***	0.121***
	(0.009)	(0.013)	(0.012)	(0.015)	(0.015)
Family Exemption	0.014	0.070*	0.073***	0.074***	0.091***
	(0.027)	(0.040)	(0.019)	(0.026)	(0.034)
UI Benefits	0.027***	0.027***	0.024***	0.029**	0.046***
	(0.007)	(0.010)	(0.008)	(0.012)	(0.010)
Children			-0.033**	-0.027*	0.016***
			(0.013)	(0.015)	(0.004)
Pre 2008 x Threshold					0.001
					(0.006)
Pre 2008					-0.050***
					(0.005)
Constant	0.204***	0.182***	0.101***	0.098***	0.095***
	(0.014)	(0.017)	(0.014)	(0.017)	(0.007)
Observations	352,197	141,864	225,423	90,359	148,029
R-squared	0.166	0.171	0.174	0.163	0.040

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 6.B

DID Within-Variation (SNAP Participation Equation 5)

VARIABLES	(1) UF. Gen.	(2) Gen. Stock	(3) UF. ABAWD	(4) ABAWD Stock	(5) ABAWD Comp.
Over Age Threshold	0.029***	0.016**	0.004	0.004	0.004
	(0.008)	(0.008)	(0.006)	(0.006)	(0.006)
Low Ex. x Threshold	-0.034***	-0.019**	-0.009	-0.007	
	(0.009)	(0.008)	(0.007)	(0.007)	
Disabled	0.030***	0.034***	0.030***	0.047***	0.032***
	(0.006)	(0.009)	(0.009)	(0.014)	(0.012)
Family Exemption	0.038**	0.080***	0.000	0.002	-0.001
	(0.015)	(0.023)	(0.014)	(0.026)	(0.020)
UI Benefits	0.009	0.012	0.005	0.012	0.008
	(0.005)	(0.007)	(0.004)	(0.007)	(0.008)
Children			0.011	0.027***	0.017***
			(0.007)	(0.010)	(0.006)
Pre 2008 x Threshold					-0.005
					(0.007)
Constant	0.082***	0.086***	0.039***	0.028**	0.032***
	(0.005)	(0.010)	(0.007)	(0.011)	(0.009)
Observations	352,197	141,864	225,423	90,359	148,029
R-squared	0.008	0.009	0.003	0.006	0.004
Number of u_lgtkey	16,011	8,459	11,945	5,937	9,526
Individual FE	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 7.A

	DID Between-Variation (Employment Equation 5)				
	(1)	(2)	(3)	(4)	(5)
VARIABLES	UF. Gen.	Gen. Stock	UF. ABAWD	ABAWD Stock	ABAWD Comp.
Over Age Threshold	-0.011***	-0.019***	-0.019***	-0.005	-0.000
	(0.003)	(0.004)	(0.006)	(0.007)	(0.007)
Low Ex. x Threshold	-0.089***	-0.035***	-0.010	-0.011	
	(0.007)	(0.008)	(0.009)	(0.011)	
Low Exemptions	0.320***	0.268***	0.098***	0.073***	
	(0.011)	(0.014)	(0.009)	(0.012)	
Disabled	-0.451***	-0.498***	-0.766***	-0.784***	-0.696***
	(0.010)	(0.013)	(0.007)	(0.010)	(0.012)
Family Exemption	-0.431***	-0.413***	-0.591***	-0.584***	-0.654***
	(0.019)	(0.036)	(0.026)	(0.039)	(0.035)
Children			0.084***	0.064***	0.013**
			(0.006)	(0.010)	(0.006)
Pre 2008 x Threshold					-0.012
					(0.009)
Pre 2008					0.035***
					(0.009)
Constant	0.440***	0.497***	0.752***	0.768***	0.784***
	(0.014)	(0.018)	(0.011)	(0.015)	(0.010)
Observations	352,192	141,863	225,423	90,359	148,023
R-squared	0.409	0.419	0.520	0.512	0.152

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 7.B

	DID Within-Variation (Employment Equation 5)				
	(1)	(2)	(3)	(4)	(5)
VARIABLES	UF. Gen.	Gen. Stock	UF. ABAWD	ABAWD Stock	ABAWD Comp.
Over Age Threshold	0.039***	0.026***	0.000	0.011	0.003
	(0.006)	(0.006)	(0.007)	(0.008)	(0.009)
Low Ex. x Threshold	-0.047***	-0.051***	-0.006	-0.019**	
	(0.007)	(0.007)	(0.010)	(0.010)	
Disabled	-0.303***	-0.326***	-0.438***	-0.457***	-0.453***
	(0.011)	(0.015)	(0.016)	(0.024)	(0.018)
Family Exemption	-0.286***	-0.322***	-0.355***	-0.391***	-0.387***
	(0.019)	(0.030)	(0.025)	(0.035)	(0.033)
Children			0.003	-0.015	0.003
			(0.009)	(0.012)	(0.010)
Pre 2008 x Threshold					-0.002
					(0.012)
Constant	0.634***	0.676***	0.816***	0.840***	0.873***
	(0.009)	(0.015)	(0.012)	(0.023)	(0.016)
Observations	352,192	141,863	225,423	90,359	148,023
R-squared	0.067	0.092	0.087	0.102	0.088
Number of u_lgtkey	16,011	8,459	11,945	5,937	9,526
Individual FE	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

8 Discussion

The regression discontinuities largely did not find any effect of treatment (aging out of the work requirements). Two specifications provide some indication that turning 60 is associated with a discontinuous decrease in employment. While these are not particularly robust, it is possible that a lack of precision may have contributed to this. As expected, the 2WFE regressions did not find any effect on employment or program participation for either work requirement. The assumption of parallel trends for participation was less convincing in the stock samples (though the ABAWD 2008 counterfactual group is great), while for employment it was less convincing in the combined samples. I disregarded the combined General sample for employment due to a failure of parallel trends. The only DID regression finding change in participation for ABAWDs was not very significant and wasn't really a convincing parallel trend so I disregard it. Removing the General Requirement appeared to increase program participation in the between regression, while it decreased program participation by an enormous amount in the regressions identified on within variation. For the former, the effects were not entirely robust and appear to be attributable to changes in participation in the comparison group, and thus are hard to credibly interpret as being the result of treatment. The latter results merit some discussion. Running these within-variation regressions separately in the high and low exemption groups (after removing the group and interaction terms) indicates that participation increases for the comparison groups after 60, while it stays relatively constant for the treatment groups (see appendix reference for these regressions). While none of these effects are statistically significant on their own, in combination they indicate that the DID interaction terms, representing the differential effects of treatment, are largely driven by changes in participation among the comparison groups. Thus, it is dubious that any significant effects are the result of treatment. Finally, the removal of the General Work Requirement in the stock sample is associated with a significant decrease in employment for both within and between regressions. Rerunning the regressions within the individual subgroups, employment in the treatment group falls by a statistically significant margin while staying constant in the comparison group, indicating that the DID effects on employment are driven by changes in the treatment group. Taken together these results tell us little about the effects of the ABAWD requirement, or the relationship between the General Requirement and program participation, but they provide tentative evidence that removing the General Work Requirement decreases employment. However, they provide tentative evidence that the removal of the General Requirement may reduce employment by 1 to 5.1 percentage points, potentially indicating that the requirement acts to increase employment. These results are strongest in DID specifications which estimate the change in employment at 60 among those subject to the requirement relative to those not subject to it.

These conclusions are broadly supported by my robustness checks. Informed partially by Cuffey et al. 2022 and Ritter 2018, I constructed alternative samples based on low education in order to evaluate the robustness of my estimates. I reran equations 1 through 5 in these samples. The results, as well as more explanation of these samples, can be found in Appendix Tables A1 through A5. The RD specifications did not differ in their effects on participation. Regarding employment, equation 1 still yields a decrease in employment in the stock General sample, equation 3 lost any significance in the combined General sample. Several regressions in several samples gained statistical significance indicating that removing the ABAWD and General Requirements increased employment. This is surprising given that this conflicts with theory, and past literature. Overall, this undermines confidence in the regression discontinuities' ability to identify the effect of treatment on the treated. The 2WFE models persist in not finding any significant effects. The DID between-variation regression's estimates to increase participation are not robust to the alternative samples. However, the decrease in participation at 60 using within-variation persisted. Perhaps most importantly, aging out of the General Requirement still significantly decreased employment in both the between and within regressions.

There are several reasons to doubt that the removal of the General Requirement causes the decrease in employment at 60. The DID estimates are premised on the assumption that those who were frequently exempt from the requirement represent a good counterfactual group and that relative to them, in the absence of treatment, employment among those with low exemptions would have changed in a similar manner as they aged. While the pretends for these groups were not statistically significantly different from one another in the stock sample, this is not a sufficient condition for parallel trends to hold. Among those who were exempt from the General Requirement more than 70% of the time before 60, 94.3% of their person-months observations had a work preventing disability, compared to 1.7% among those who were exempt $< 30\%$ of the time. Correspondingly, only 1.6% of the high exemption group's pre-treatment observations were employed, compared to 76.2% for the low exemption treatment group. The largest estimated effect indicates an employment decrease of 5.1 percentage points relative to the comparison group, but the comparison group's initial employment was so low as to make a commensurate decrease impossible. Given this, it seems plausible that the high and low exemption groups may have systematically different potential outcomes. However, as shown in figure 4.B.2, the high exemption group experiences an upwards trend in employment after 60, and this is partially responsible for the estimated relative decrease in employment among the low exemption group.

Additionally, the methods of this paper suffer from several serious issues which may have contributed to the lack of many significant results. Most of the known first order issues discussed throughout this paper would cause results to be biased towards 0, potentially explaining the lack of clear effects where other papers found them. An inability to observe or account for waivers and 15% exemptions to the ABAWD Work Requirement means that a significant proportion of ABAWDS in our sample were not subject to the requirement, even before they turned 50. This would dilute our treatment group, causing severe downwards bias and limiting the estimated effects of removing the ABAWD requirement to ITT at best. Thus we would not expect to observe the removal of the ABAWD requirement to have much effect on employment or SNAP participation. This likely significantly explains why I found no effect of the ABAWD requirement while past papers did. Additionally, if the doughnut design fails to account for anticipatory effects then this would move the average pretreatment value of the dependent variable in the direction of the true effect, thus making it appear as if treatment had less effect. The samples in this paper likely also suffer from severe sample dilution, including a large number of people for whom SNAP and the work requirements were not relevant, as evidenced by the fact that a very small proportion of each sample participating in food stamps, both before and after the removal of work requirements.(turn this into a good thing bc it indicates that other papers may have been way off and most of the people in their samples on food stamps might have not been under the ABAWD requirement afterall). Finally, dynamic treatment effects are likely the most intractable source of downwards bias in this paper. As discussed previously, most of the mechanisms through which the removal of the work requirements would impact employment and participation would take place gradually over time. People may not immediately be aware of the change in program rules on their 50th or 60th birthdays. New applicants may not begin receiving benefits for months. The effects of a reduced rate of program attrition will only manifest cumulatively over months or years. While the doughnut design may account for some of this, it is likely that my RD and DID specifications would fail to capture most of the dynamic effects, thus making the effects of removing work requirements appear to be smaller than they really are. In an attempt to partially account for this, I estimated a variant of the DID models which included a series of age-in-months dummies, though this did not meaningfully alter any effects of interest or their significance.

Considering these points, there are reasons to think that the estimates would have been close to 0 even if the true treatment effects were not. Several of these issues are common to most of the papers on SNAP's work requirements, as described in my literature review. However, the severe dilution caused by the inability to account for ABAWD waivers is an issue specific to the present study. This may explain why I fail to demonstrate the effects on treatment and employment observed by Cuffey et al. 2022 and Gray et al. 2023.

However, several potential issues likely inflate the apparent effects of work requirements, biasing my effects away from 0. As discussed in the theory section, turning 60 itself may impact people's behavior with regards to my outcomes of interest, and these effects would be conflated with the effect of removing the General Work Requirement. This would globally bias the effects on participation in a positive direction and the effects on employment in the negative direction. Of particular concern, the tendency to see a big 60 on your birthday cake and decide that you've worked long enough and that it's time to retire can only occur among those who were working in the first place. Thus this will primarily only occur in the low exemptions group and not in the high exemptions group. Additionally, there may be several contemporaneous SNAP policy shocks at 60, though it is unclear the extent to which they are likely to be an issue. Removal of the gross income requirement at 60 could be a confounding factor, causing upwards bias in the participation response. However, this would primarily affect those earning too much to receive more than the token minimum SNAP benefits, and thus might not induce significant program entry. More concerning is the possibility of discontinuous changes in certification time, outreach, and E&T at ages 50 or 60. While some of these could theoretically be considered part of the work requirement, others would cause bias away from 0. However, just as with the dynamic treatment effects, most of these would manifest gradually over time, and thus the RD and DID regressions would evade a portion of this bias. Point out that other papers also have most of these issues. Perhaps most seriously, I cannot distinguish between dynamic treatment effects and the effects of age. To the extent that an outcome is different post treatment, a before and after comparison would encompass both these effects, thus appearing to make the removal of work requirements appear more impactful than they are. It is for this reason that I do not attempt to capture the dynamic treatment effects using an event study design.

9 Conclusion

The ways in which SNAP's work requirements affect the livelihood of low-income Americans is a subject of much debate. Do more stringent requirements create a greater incentive toward self-sufficiency through work, or simply deprive poor people of a resource that could help them achieve stability? The answer to this question is vital to determining the best path forward for policy. This paper attempts to answer this question by examining the effect of SNAP's work requirements on employment and program participation among older adults. I use DID and RD specifications to estimate the effects of removing the ABAWD Requirement and General Requirement at 50 and 60.

Overall, I do not find unambiguously credible evidence that either work requirement has an effect on employment or SNAP participation. Many methodological problems limit my ability to paint a clear picture of the requirements' impacts. In particular, severe sample dilution prevents me from saying anything meaningful about the ABAWD requirement. When the General Requirement is removed at 60, the probability of employment decreases by 1-5.1 percentage points, and this appears to be driven by individuals who were subject to the requirement. This provides tentative evidence that the General Requirement may encourage employment among adults in their late 50s, but it would be unreasonable to accept this conclusion at face value. These results are not entirely robust to alternate specifications, there are serious threats to internal validity, and I cannot rule out the possibility that this is simply the effect of turning 60. Further research will be needed before we can say anything definitive about the effects of the General Work Requirement.

I identify several lessons and areas for improvement in further research. In particular, this study, like most existing studies on the ABAWD Requirement (Cuffey et al. 2022, Gray et al. 2023, Han 2022, Ritter 2018), is likely biased by dynamic treatment effects and/or anticipatory effects. However, I make several improvements on past methods, including focusing on observing individuals over time in order to avoid certain kinds of sample selection. This paper also identifies and partially accounts for several previously neglected interactions between SNAP's work requirements. Finally, I construct the first theoretical model of the effects of the General Work Requirement.

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